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لم يتم حل الاسايمنت التالت

Formative 1 answered

1	**Which stage of disease progression involves the highest severity of signs and symptoms?**
A	Decline stage with reduced pathogen load
B	Incubation stage with silent pathogen growth
C	Convalescence stage with gradual recovery
D	Illness stage with pronounced clinical features

2	**A newborn receives maternal antibodies through the placenta, protecting him against infections during the first months of life. Which type of immunity is this?**		
A	Artificial immunity	B	Specific resistance
C	Passive immunity	D	Active immunity

3	**Which definition best distinguishes disease?**
A	A social inability to maintain daily roles
B	A mild subjective discomfort without signs
C	A physiological or psychological dysfunction
D	A temporary response to emotional stress

4	**In typhoid, the time during which the organism can be transmitted is known as what?**		
A	Period of communicability	B	Virulence
C	Pathogenicity	D	Secondary attack rate

5	**Which stage of natural disease history occurs before symptom onset but after pathological changes begin?**
A	Convalescence period after active illness
B	Decline period during pathogen elimination
C	Preclinical stage representing biological onset
D	Prodromal stage with nonspecific symptoms

6	**A man is bitten by a rabid dog and develops rabies; what direct transmission mode is involved?**		
A	Soil contact	B	Vertical transmission
C	Droplet infection	D	Contact with animals

7	**Which statement defines illness?**
A	A permanent decline from normal health
B	A subjective feeling of not being well
C	A socially assigned disability state
D	A measurable biological malfunction

8	**Case Fatality Ratio (CFR) is a good measure of?**						
A	Pathogenicity	B	Infectivity	C	Virulence	D	Infectiousness

9	**Which period is typically associated with maximum clinical severity?**		
A	Decline period	B	Incubation period
C	Convalescence period	D	Illness period

10	**A student with mumps is isolated; his apparently healthy brother is kept at home for two weeks. What is the brother's carrier state?**						
A	Temporary	B	Convalescent	C	Incubatory	D	Healthy

11	**Which statement best reflects the spectrum of health?**
A	Illness always indicates irreversible decline
B	Health ranges from complete wellness to death
C	Disease and health exist as fixed categories
D	Health begins only when symptoms appear

12	**Malaria parasite multiplies and develops inside a mosquito; which type of transmission is this?**		
A	Cyclodevelopmental	B	Propagative
C	Cyclopropagative	D	Mechanical

13	**A patient receives hepatitis B vaccine produced using recombinant DNA technology. Which type of vaccine is this?**		
A	Recombinant	B	Inactivated
C	Live attenuated	D	Subunit

14	**A patient inhales droplets 5–10 µm from a measles case; what type of transmission is this?**		
A	Vehicle	B	Droplet infection
C	Air borne	D	Vector

15	**A parasite develops but does not multiply inside a vector; what type of biological transmission is this?**		
A	Mechanical	B	Cyclodevelopmental
C	Propagative	D	Cyclopropagative

16	**The occurrence of health or disease depends on which interaction among?**
A	Agent, environment and host
B	Exposure, induction and disease
C	Time, place and person
D	Implementation, monitoring and evaluation

17	**A patient receives chloroquine to prevent malaria. Which type of prophylaxis is this?**		
A	Chemoprophylaxis	B	Seroprophylaxis
C	Active immunization	D	Passive immunization

18	**A community living near industrial pollution reports increased respiratory illnesses. Which determinant of health is most implicated?**
A	Cultural environment based on local traditions
B	Genetic predispositions with inherited abnormalities
C	Physical environment associated with air quality exposure
D	Social networks with variable interpersonal relations

19	**A child receives Salk polio vaccine prepared from killed organisms. Which type of vaccine is this?**		
A	Recombinant	B	Subunit
C	Live attenuated	D	Inactivated

20	**Which period of disease is characterized by pathogen multiplication with vague symptoms?**		
A	Decline stage	B	Illness stage
C	Prodromal stage	D	Convalescence stage

21	**A reconstituted measles vaccine is left unused for several hours. What happens?**		
A	Potency increased	B	Potency unaffected
C	Potency maintained	D	Potency lost

22	**A hospital sterilizes surgical instruments to prevent infections. Which preventive measure is this?**		
A	Secondary prevention	B	Rehabilitation
C	Specific protection	D	Health promotion

23	**Infected dust >10 µm commonly spreads which disease in hospitals?**						
A	Smallpox	B	Tuberculosis	C	Measles	D	Influenza

24	**Which dimension of health is most closely reflected by self-control and stable emotional responses?**
A	Vocational dimension emphasizing economic needs
B	Emotional dimension based on response regulation
C	Social dimension emphasizing group relationships
D	Mental dimension focusing on cognitive processes

25	**A housefly transfers typhoid organisms from feces to food via its legs; what type of vector is it?**		
A	Cyclopropagative	B	Biological
C	Propagative	D	Mechanical

26	**A cataract patient receives intraocular lens implantation to restore vision. Which aspect of prevention is this?**		
A	Secondary prevention	B	Primary prevention
C	Disability limitation	D	Rehabilitation

27	**Which determinant best explains why lower education levels correlate with poor health outcomes?**
A	Improved awareness of community disease patterns
B	Enhanced access to diagnostic and curative facilities
C	Guaranteed employment regardless of skill set
D	Increased stress and reduced confidence in health decisions

28	**Which factor belongs to the environmental component of the triad?**
A	Psychological resilience of individuals
B	Genetic predisposition to illness
C	Biological dose of the pathogen
D	Physical conditions where the host lives

29	**A patient with advanced cancer receives aggressive treatment to prevent further disability. Which level of prevention is this?**		
A	Secondary prevention	B	Tertiary prevention
C	Rehabilitation	D	Primary prevention

30	**A person develops cutaneous anthrax after handling contaminated wool; what is the portal of entry?**						
A	Skin	B	Conjunctival	C	GI tract	D	Respiratory

31	**Which determinant explains population health patterns beyond individual control?**
A	Personal preference for certain food types
B	Individual desire for physical activity
C	Contextual social and economic environment
D	Daily choice of recreational hobbies

32	**Which statement reflects the concept of well-being?**
A	A continuous state of economic success
B	A lack of biological abnormalities
C	A positive judgment of life and feeling good
D	A clinical absence of disease

33	**Which element is an example of a social determinant of health?**		
A	Number of daily exercise sessions	B	Personal taste preferences
C	Red blood cell count levels	D	Income and social status

34	**Which component of the epidemiological triad refers to host-related characteristics?**		
A	Environmental factors	B	Agent factors
C	Host factors	D	Social behavior factors

35	**Infectious organisms present on utensils represent what concept?**						
A	Infection	B	Colonization	C	Infestation	D	Contamination

36	**A patient receives human rabies immunoglobulin. Which type of prophylaxis is this?**		
A	Chemoprophylaxis	B	Active immunization
C	Passive natural immunity	D	Seroprophylaxis

37	**Which health dimension includes the ability to maintain relationships and integrate with society?**
A	Vocational dimension describing work capability
B	Spiritual dimension describing philosophical beliefs
C	Physical dimension describing organ performance
D	Social dimension describing interpersonal functioning

38	**An infected person who shows no symptoms but remains infectious for life is what type of carrier?**						
A	Incubatory	B	Chronic	C	Healthy	D	Convalescent

39	**Cholera transmitted through contaminated water represents which route?**						
A	Vehicle	B	Vector	C	Air borne	D	Fomite

40	**A young couple seeks premarital counseling to avoid genetic disorders in their offspring. Which preventive measure is being applied?**		
A	Health education	B	Disability limitation
C	Genetic counseling	D	Rehabilitation

41	**Which determinant explains why employed people often have better health outcomes?**
A	Access to entertainment facilities
B	Having predictable daily routines
C	Working longer hours without breaks
D	Better working conditions and job control

42	**A person exposed to influenza but protected due to existing antibodies is showing what factor?**		
A	Communicability	B	Virulence
C	Pathogenicity	D	Host immunity

43	**A patient inhales droplet nuclei <5 µm from a TB case; what mode of transmission is this?**		
A	Vehicle transmission	B	Fomite transmission
C	Air borne transmission	D	Droplet infection

44	**A worker recovering from a back injury is given modified work conditions to safely return to employment. Which preventive measure is this?**		
A	Rehabilitation	B	Tertiary prevention
C	Secondary prevention	D	Primary prevention

45	**A child with mild respiratory infection is scheduled for vaccination. Should vaccination be postponed?**		
A	Yes, only for OPV	B	No, not contraindicated
C	Yes, only for BCG	D	Yes, contraindicated

46	**A food handler remains infectious with typhoid for years without symptoms; what type of carrier is he?**						
A	Incubatory	B	Convalescent	C	Chronic	D	Temporary

47	**A new influenza strain spreads across continents; what level of occurrence is this?**						
A	Endemic	B	Pandemic	C	Epidemic	D	Outbreak

48	**A newborn acquires ophthalmia neonatorum while passing through an infected birth canal; what is the portal of entry?**		
A	Skin	B	Conjunctiva
C	GI tract	D	Respiratory tract

49	**Dengue virus multiplies inside Aedes mosquitoes; what type of transmission is this?**		
A	Propagative	B	Cyclodevelopmental
C	Cyclopropagative	D	Mechanical

50	**A patient with hearing loss is provided with a hearing aid to restore communication ability. This is an example of:**		
A	Specific protection	B	Disability limitation
C	Health promotion	D	Rehabilitation

51	**Which description characterizes the agent in the epidemiological triad?**
A	The cultural behaviors of communities
B	The surroundings affecting disease spread
C	The factor responsible for causing disease
D	The internal immunity of individuals

52	**How easily disease spreads among susceptible contacts is measured by which indicator?**		
A	Pathogenicity	B	Case fatality rate
C	Period of communicability	D	Secondary attack rate

53	**Soil is a reservoir for which disease?**						
A	Histoplasmosis	B	Measles	C	Rabies	D	Dengue

54	**A patient with impairment due to spinal injury is unable to perform routine activities. Which stage of disease progression is this?**						
A	Impairment	B	Handicap	C	Disability	D	Disease

55	**A patient recovering from stroke is enrolled in occupational therapy to regain independence. Which level of prevention is this?**		
A	Primary prevention	B	Tertiary prevention
C	Secondary prevention	D	Rehabilitation

56	**A patient recovering from influenza develops immunity that lasts only a short period. Which type of immunity is this?**						
A	Absolute	B	Moderate	C	Solid	D	Transient

57	**Plague bacilli multiply inside a rat flea; what is this?**		
A	Propagative transmission	B	Mechanical transmission
C	Cyclodevelopmental transmission	D	Cyclopropagative transmission

58	**A disease that produces lifelong immunity likely has which characteristic?**		
A	High virulence	B	Strong antigenic power
C	Weak antigenic power	D	Low pathogenicity

59	**Which factor most influences the length of an incubation period?**
A	Dose, site of infection, and host immunity level
B	Personal preferences, habits, and attitudes
C	Age, ethnicity, and family health history
D	Social networks, occupation, and culture

60	**A patient feels unwell, yet clinical exams show no abnormality. This state is best classified as:**
A	Sickness as social role impairment in society
B	Pathology as an advanced stage of organ damage
C	Disease as a measurable physiological dysfunction
D	Illness as a subjective experience of discomfort

61	**A child with neurological disorder should not receive which vaccine component?**						
A	Tetanus	B	Pertussis	C	Hepatitis B	D	Measles

62	**A child receives the measles vaccine and develops immunity without contracting the disease. Which type of immunity is this?**		
A	Passive immunity	B	Natural active immunity
C	Non-specific resistance	D	Artificial active immunity

63	**Which physical environment factor contributes positively to health?**
A	Access to safe water and clean air
B	Residence in overcrowded housing
C	Work in hazardous industrial sites
D	Constant exposure to excessive noise

64	**A child receives vitamin A supplementation at 18 months. Which dose is given?**						
A	200,000 IU	B	100,000 IU	C	50,000 IU	D	300,000 IU

65	**A patient with tuberculosis is treated promptly to reduce illness duration and prevent spread in the community. Which principle of prevention is applied?**
A	Health promotion
B	Early diagnosis and prompt treatment
C	Rehabilitation
D	Disability limitation

66	**Which component of the epidemiological triad involves characteristics of the disease-causing entity?**
A	Environmental factors describing physical conditions
B	Behavioral factors describing coping strategies
C	Host factors describing personal susceptibility
D	Agent factors describing pathogen characteristics

67	**Which phase of natural disease history involves biological changes without symptoms?**		
A	Outcome phase	B	Recurrence phase
C	Clinical phase	D	Preclinical phase

68	**A patient with impairment in leg function is unable to walk, resulting in disability. Which stage of disease progression does this represent?**						
A	Impairment	B	Disease	C	Handicap	D	Disability

69	**A patient receives influenza vaccine prepared from a viral component. Which type of vaccine is this?**						
A	Subunit	B	Recombinant	C	Toxoid	D	Inactivated

70	**A 45-year-old woman undergoes a mammogram to detect breast cancer before symptoms appear. This is an example of:**		
A	Secondary prevention	B	Tertiary prevention
C	Rehabilitation	D	Primary prevention

71	**Which definition best reflects the WHO concept of health?**
A	A personal perception of happiness and satisfaction
B	A simple absence of disease and clinical infirmity
C	A complete state of physical, mental, and social well-being
D	A stable physiological state with minimal pathology

72	**A newborn receives BCG vaccine at birth. Which route is used?**						
A	Oral	B	SC	C	IM	D	ID

73	**A patient receives rabies vaccine prepared in human diploid cells. Which type of vaccine is this?**		
A	Recombinant vaccine	B	Tissue culture vaccine
C	Subunit vaccine	D	Inactivated vaccine

74	**A community health program emphasizes nutrition education, personal hygiene, and lifestyle modification to strengthen individuals against disease. Which type of intervention does this represent?**		
A	Specific protection	B	Disability limitation
C	Health promotion	D	Rehabilitation

75	**A patient recovering from amoebiasis continues to shed cysts; what type of carrier is this?**						
A	Convalescent	B	Chronic	C	Temporary	D	Incubatory

76	**A patient with advanced disability is given vocational training to restore earning capacity. Which aspect of rehabilitation is this?**		
A	Psychological rehabilitation	B	Vocational rehabilitation
C	Social rehabilitation	D	Physical rehabilitation

77	**Leptospirosis consistently maintained by rodents in a region represents what?**						
A	Sporadic	B	Epidemic	C	Pandemic	D	Endemic

78	**The ability of a microorganism to cause disease is called what?**		
A	Virulence	B	Communicability
C	Pathogenicity	D	Antigenic power

79	**A country with constant low-level malaria transmission shows what pattern?**						
A	Sporadic	B	Epidemic	C	Endemic	D	Pandemic

80	**A school introduces daily physical activity and healthy eating habits for all students, regardless of their current health status. Which strategy of primordial prevention is being applied?**		
A	High-risk strategy	B	Disability limitation
C	Population strategy	D	Rehabilitation strategy

Formative 2

1	**Which of the following is the best method for disposal of hospital refuse such as syringes and needles?**		
A	Incineration	B	Burial in soil
C	Washing with water	D	Storing in sealed bags

2	**In a densely populated urban district, tuberculosis incidence has remained stable for 5 years despite good treatment coverage. Which factor most likely explains the continued transmission?**
A	Low case fatality rate
B	High proportion of extrapulmonary TB cases
C	Delayed diagnosis of infectious pulmonary cases
D	High BCG vaccination coverage

3	**Which clinical symptom combination most strongly suggests pulmonary TB?**
A	Chronic cough above 3 weeks with fever and weight loss
B	Persistent headache with photophobia and vomiting
C	Acute diarrhea with rapid dehydration and cramps
D	Sudden joint pain with localized swelling and redness

4	**Which preventive measure offers partial protection against leprosy?**		
A	MMR vaccine	B	BCG vaccine
C	Polio vaccine	D	Hepatitis B vaccine

5	**A public health survey finds that tuberculosis prevalence is highest in low-income urban settlements. Which determinant most strongly explains this distribution?**		
A	Seasonal climate changes	B	Genetic susceptibility
C	High vaccination coverage	D	Overcrowding and poor ventilation

6	**Which vaccine type offers improved immunogenicity and herd protection against meningococcal disease?**
A	Conjugate vaccines targeting A, C, W-135, Y
B	DNA-based vaccines targeting multiple serotypes
C	Live attenuated vaccines targeting serogroup B
D	Polysaccharide vaccines with lifelong protection

7	**In developing countries, the prevalence of HAIs is significantly higher than in developed countries. What is the approximate prevalence range in developing countries?**						
A	50–60%	B	15–20%	C	25–40%	D	7–12%

8	**Which environmental control is most effective during meningitis epidemics?**
A	Encouraging outdoor exposure during peak hours
B	Increasing humidity in enclosed environments
C	Reducing access to shared recreational facilities
D	Improving ventilation in crowded institutions

9	**Which statement describes the role of isolation during meningococcal epidemics?**
A	It has limited effect because carriers are more significant sources
B	It replaces the requirement for vaccination campaigns
C	It is essential because clinical cases transmit the majority
D	It eliminates the need for chemoprophylaxis programs

10	**Which regimen is recommended for preventive therapy in TB contacts?**
A	Monthly streptomycin injections
B	Daily isoniazid at adult dose of 300 mg
C	Weekly rifampicin at minimal dosing
D	Quarterly ethambutol dosing

11	**Which leprosy reaction is commonly seen in lepromatous leprosy?**		
A	Reversal reaction	B	Erythema nodosum leprosum (ENL)
C	Type IV hypersensitivity	D	Anaphylactic reaction

12	**Which population serves as the main reservoir for M. leprae?**						
A	Rodents	B	Humans	C	Soil bacteria	D	Birds

13	**A close contact of an infectious TB case tests negative on PPD during exposure and returns after 10 weeks for repeat testing. Why is this necessary?**
A	To verify improvement of physical symptoms post-exposure
B	To assess early lung damage following acute exposure
C	To detect delayed hypersensitivity conversion after exposure
D	To confirm eradication of infection by immune system

14	**Which factor most significantly affects TB transmission probability?**
A	Type of food and nutritional habits of contacts
B	Genetic similarities between index and contacts
C	Age, gender, and education of exposed persons
D	Infectiousness, environment, duration of exposure

15	**In a rural community, the prevalence of tuberculosis is decreasing but the incidence remains unchanged. Which interpretation is most appropriate?**
A	Population immunity is declining
B	Duration of disease is shortening due to improved treatment
C	Case detection is decreasing
D	Transmission is increasing

16	**How soon after beginning effective antibiotic therapy does a typical meningococcal case stop being infectious?**
A	After the completion of the entire drug regimen
B	Only after laboratory confirmation of sterile CSF
C	Within the first 24 hours of treatment initiation
D	Within the first 5 days after clinical improvement

17	**A patient develops infection after lumbar puncture due to poor aseptic technique. Which cause of HAI is most relevant?**
A	Drug resistance
B	Contaminated food
C	Overcrowding
D	Failure to observe aseptic precautions

18	**What is the primary mode of transmission of leprosy?**
A	Direct skin contact
B	Respiratory droplets during prolonged close exposure
C	Insect bites
D	Contaminated water

19	**A patient develops infection after being treated with contaminated wound dressings. Which transmission mechanism is most likely?**		
A	Droplet infection	B	Fomite transmission
C	Contact transmission	D	Dust infection

20	**Universal blood and body fluid precautions are primarily aimed at preventing transmission of which infections?**		
A	Escherichia coli and Klebsiella	B	Tuberculosis and malaria
C	HBV and HIV	D	Influenza and measles

21	**A hospital maintains 12 ft spacing between beds and ensures cross ventilation. Which transmission route is being controlled?**		
A	Contact transmission	B	Droplet transmission
C	Dust transmission	D	Fomite transmission

22	**Which finding best differentiates lepromatous from tuberculoid leprosy in a field epidemiology survey?**
A	Strong lepromin positivity
B	Multiple symmetrical lesions with nasal mucosa involvement
C	Localized lesions with sharp margins
D	Presence of fever

23	**Which mass chemoprophylaxis policy applies in meningitis epidemics?**
A	Recommended for all individuals in open community settings
B	Recommended solely for symptomatic clinical cases
C	Recommended only for closed, medically supervised communities
D	Recommended only after vaccination has been completed

24	**Over 60% of global deaths are attributed to NCDs. Which disease group contributes the highest mortality?**		
A	Digestive diseases	B	Musculoskeletal disorders
C	Cardiovascular diseases	D	Chronic respiratory diseases

25	**Which group is MOST vulnerable to HAIs?**		
A	Vaccinated individuals	B	Post-surgery patients
C	Hospital staff	D	Healthy adults

26	**Which modifiable risk factor contributes to approximately 3.3 million deaths annually worldwide?**		
A	Harmful use of alcohol	B	Insufficient physical activity
C	Tobacco use	D	High salt intake

27	**Which measure best reflects the burden of chronic diseases beyond mortality figures?**
A	Years of Potential Life Lost (YPLL)
B	Case fatality rate
C	Attack rate
D	Incidence rate

28	**A meningitis epidemic is declared in a refugee camp. Which public health measure is most appropriate?**
A	Replacing all camp healthcare staff immediately
B	Blocking all transportation routes to nearby towns
C	Enhancing ventilation and reducing crowding in shelters
D	Increasing surface disinfection of shared objects

29	**Which factor most contributes to rapid spread during meningitis epidemics?**
A	High humidity and frequent exposure to rainfall
B	High survival of the organism outside the human body
C	Minimal contact between individuals in communities
D	Overcrowded living environments with poor ventilation

30	**Which major component makes DOTS an effective TB control strategy?**
A	Sole reliance on radiographic screening programs
B	Exclusive treatment of smear-negative patients
C	Direct observation of therapy during initial months
D	Mandatory hospitalization of all infectious cases

31	**A region experiences a sudden increase in tuberculosis incidence after several years of decline. Which situation most likely contributed to this change?**
A	Decreased population density
B	Increase in immunocompromising conditions such as HIV/AIDS
C	Expansion of vaccination programs
D	Improved socioeconomic conditions

32	**Why is rifampicin considered the most crucial drug in multidrug therapy (MDT) for leprosy control?**
A	It rapidly kills M. leprae within days, reducing infectiousness
B	It prevents nerve thickening
C	It improves skin pigmentation
D	It is the cheapest drug

33	**What is the average incubation period for leprosy?**						
A	1 year	B	5 years	C	20 years	D	10 years

34	**A country reports that most new tuberculosis cases occur among young adults aged 20–40 years. Epidemiologically, this pattern most likely indicates:**
A	Reactivation of old infections in the elderly
B	Active and ongoing transmission in the community
C	Effective childhood vaccination programs
D	Declining TB transmission

35	**Which major risk period follows TB infection and carries the highest likelihood for clinical disease development?**
A	First 6–12 months following initial infection
B	Second decade of life for all age groups
C	Final months of convalescence after therapy
D	Fifth to eighth year after initial exposure

36	**Which clinical feature is most strongly associated with early invasive meningococcal disease progression?**
A	Chronic fatigue with reduced appetite
B	Gradual muscle weakness over several weeks
C	Persistent dry cough with low-grade fever
D	Severe headache with photophobia and vomiting

37	**During contact investigation of a smear-positive tuberculosis patient, several household contacts are found to have latent infection but no active disease. Which control measure is most appropriate to prevent future cases?**		
A	Isolation for 2 weeks	B	Preventive chemotherapy
C	Repeat sputum microscopy monthly	D	Immediate hospitalization

38	**A university dormitory reports several students with nasopharyngitis followed by one suspected meningitis case. What is the most important mode of transmission to consider?**
A	Fomite contamination through shared equipment
B	Vector-borne spread from local insect populations
C	Waterborne exposure in shared sanitation areas
D	Airborne spread through prolonged close contact

39	**Which action is most vital during meningitis epidemic control efforts?**
A	Delaying treatment until laboratory confirmation
B	Minimizing surveillance to reduce public concern
C	Replacing antibiotics with environmental modification
D	Early case detection and prompt initiation of treatment

40	**Which serogroups of Neisseria meningitidis are most commonly associated with invasive disease worldwide?**						
A	B, C, X, Y	B	A, E29, X, Y	C	A, B, C, W-135	D	W-135, X, Y, Z

41	**A hospital outbreak of Klebsiella infections is traced back to contaminated IV fluids. Which transmission mechanism is most likely?**		
A	Contact transmission	B	Dust infection
C	Iatrogenic transmission	D	Droplet infection

42	**A patient with untreated multibacillary leprosy lives with several family members. Which group is most at risk of developing infection?**
A	People living in neighboring villages
B	Household contacts with prolonged exposure
C	Health workers using protective measures
D	Casual visitors

43	**Which administrative measure is essential for outbreak detection and control of HAIs?**
A	Hospital Infection Control Committee
B	Concurrent disinfection
C	Barrier nursing
D	Vaccination of patients

44	**A hospital staff member with influenza continues working and infects multiple patients. Which preventive measure was neglected?**		
A	Barrier nursing	B	Concurrent disinfection
C	Staff with infections off duty	D	Task nursing

45	**Raised cholesterol is most strongly associated with which outcome?**		
A	Chronic respiratory disease	B	Stroke and heart disease
C	Musculoskeletal disorders	D	Diabetes

46	**A region reports low prevalence but continues active surveillance and early treatment programs. What is the most important reason for maintaining these measures?**
A	To shorten treatment duration
B	To increase hospital admissions
C	To eliminate all bacterial diseases
D	To prevent reintroduction and continued transmission

47	**In a hospital where overcrowding is common and aseptic precautions are poorly followed, which factor most strongly predisposes to HAIs?**		
A	Drug resistance in organisms	B	Adequate ventilation
C	Vaccination of staff	D	Proper sterilization of equipment

48	**During disaster or war situations, interruption of leprosy treatment programs may lead to which major public health consequence?**
A	Immediate eradication of the disease
B	Shorter incubation period
C	Reduction in new infections
D	Increased transmission and complications

49	**Which screening measure is routinely recommended for all close TB contacts?**
A	PPD tuberculin skin testing for initial evaluation
B	Routine blood culture to detect bacteremia
C	Mandatory CT scan for early lung imaging
D	Immediate sputum culture for every exposed person

50	**A patient admitted for pneumonia develops urinary tract infection after catheterization. Which source of infection is most likely?**		
A	Contaminated food	B	Environmental reservoir
C	Cross-infection from staff	D	Patient's own flora

51	**Which indirect cost is commonly associated with chronic diseases?**		
A	Productivity loss	B	Medication expenses
C	Hospital bills	D	Surgery costs

52	**Why might borderline leprosy shift toward the lepromatous end of the spectrum if untreated?**
A	Due to reinfection with a different strain
B	Because it is immunologically unstable with declining cell-mediated immunity
C	Due to improved immunity
D	Due to failure of B-cell activation

53	**A community health worker identifies a patient with hypopigmented skin lesions and loss of sensation. Early treatment is started immediately. What is the most important public health benefit of this action?**
A	Prevention of nerve damage and disability
B	Shortening of the incubation period
C	Elimination of environmental sources of infection
D	Reduction of stigma in the community

54	**A tuberculosis control program identifies that most transmission occurs in overcrowded households. Which primary prevention strategy would most effectively reduce this risk?**
A	Expansion of BCG vaccination programs
B	Increasing antibiotic prophylaxis for the population
C	Routine mass chest X-ray screening
D	Early detection and treatment of infectious cases

55	**During a meningitis epidemic, which indicator most reliably signals that the outbreak is intensifying?**
A	A consistent reduction in severe clinical cases
B	A stable carrier level across multiple months
C	A gradual decline in reported nasopharyngeal cases
D	A progressive rise in asymptomatic carrier rates

56	**A contact has a positive PPD but normal chest X-ray. What is the appropriate next step?**
A	Request repeated sputum smear daily
B	Begin isoniazid preventive treatment
C	Delay therapy until symptoms appear
D	Start full four-drug TB treatment

57	**Which environmental condition most strongly contributes to meningococcal meningitis outbreaks in the African meningitis belt?**
A	Cold and dry seasonal winds with dust exposure
B	Hot and humid equatorial rainfall patterns
C	Stable year-round temperatures with high moisture
D	Irregular monsoon cycles with mild variations

58	**What is the first-line drug for meningococcal meningitis treatment?**
A	Rifampicin used for both cases and contacts
B	Ceftriaxone reserved exclusively for prophylaxis
C	Ciprofloxacin given as a short-course regimen
D	Penicillin administered promptly on diagnosis

59	**Which group is most at risk among contacts of infectious TB cases?**
A	Occasional visitors with minimal direct interactions
B	Casual community members without close proximity
C	Close household contacts with prolonged daily exposure
D	Distant neighbors with outdoor-only interactions

60	**During a screening campaign, a patient presents with a single well-defined hypopigmented lesion with marked sensory loss and strong cell-mediated immunity. Which type of leprosy is most likely?**		
A	Lepromatous leprosy	B	Tuberculoid leprosy
C	Multibacillary leprosy	D	Borderline leprosy

61	**A nurse forgets to disinfect her hands after handling a patient's sputum and then attends another patient. Which transmission route is most likely?**		
A	Fomite transmission	B	Dust infection
C	Contact transmission	D	Droplet infection

62	**In a village with several untreated multibacillary leprosy cases, new infections continue to appear among close household contacts. Which factor most strongly explains this pattern?**
A	Seasonal climate changes
B	Contaminated water supply
C	Prolonged close exposure to infectious untreated cases
D	Genetic susceptibility of villagers

63	**Which contacts should receive preventive therapy even if initially PPD-negative?**
A	Elderly individuals without recent illness history
B	Individuals with occasional short-term exposures
C	Children under five years or HIV-infected individuals
D	Adults with stable immune systems and no symptoms

64	**A household contact tests negative on a PPD immediately after exposure. What is the next correct step?**
A	Repeat the PPD after 2–3 months for possible conversion
B	Begin full TB treatment without further evaluation
C	Discontinue follow-up due to low transmission risk
D	Initiate immediate chest radiography every week

65	**A national tuberculosis control program prioritizes early detection of infectious cases. Which group should be targeted first for screening to most effectively interrupt transmission?**
A	Patients with extrapulmonary tuberculosis
B	Patients who completed treatment years ago
C	Individuals previously vaccinated with BCG vaccine
D	Household contacts of smear-positive tuberculosis cases

66	**During an inter-epidemic period in a Sub-Saharan African region, surveillance shows a carrier rate of 30%. What does this indicate epidemiologically?**
A	A moderate baseline that may rise before potential epidemics
B	A stable prevalence with low likelihood of immediate outbreaks
C	A gradual decline toward post-epidemic recovery phase
D	A negligible level insufficient for transmission dynamics

67	**A meningitis outbreak occurs in a military training base. What is the most appropriate measure?**
A	Replace routine treatment with serologic screening only
B	Restrict antibiotic use only to confirmed clinical cases
C	Postpone all surveillance activities during the outbreak
D	Implement mass chemoprophylaxis within the base

68	**What cardinal sign confirms the diagnosis of leprosy in endemic areas?**
A	Skin lesion with definite sensory loss
B	Positive lepromin test
C	Leukocytosis
D	Elevated ESR

69	**Which statement about prevention of chronic diseases is correct?**
A	Genetic risks cannot be modified
B	Only 20% are preventable
C	Prevention is less effective than treatment
D	80% are preventable with existing measures

70	**Which diagnostic test can confirm TB before it becomes symptomatic?**
A	Tuberculin skin test indicating latent infection
B	Blood culture indicating systemic dissemination
C	Chest radiograph indicating early cavitation only
D	Sputum smear indicating only advanced disease

71	**Which preventive measure should be strengthened during early meningococcal epidemic detection?**
A	Immediate nationwide vaccination without targeting
B	Daily testing of schoolchildren for meningococcal bacteria
C	Surveillance to identify rises in nasopharyngeal carriage
D	Mass treatment of all reported upper respiratory infections

72	**A patient develops permanent disability after a spinal injury. Which characteristic of chronic disease does this illustrate?**		
A	Residual disability	B	Temporary impairment
C	Short-term supervision	D	Reversible pathology

73	**In a control program, emphasis is placed on multidrug therapy rather than single-drug treatment. What is the main reason for using combination therapy?**
A	To shorten incubation period
B	To reduce treatment cost
C	To prevent development of drug resistance
D	To improve vaccine effectiveness

74	**Which statement about vaccination during meningitis epidemics is correct?**
A	Vaccination is contraindicated in all adolescent groups
B	Conjugate vaccines covering A, C, W-135, Y are protective
C	Polysaccharide vaccines offer lifelong herd immunity
D	Vaccination replaces the need for chemoprophylaxis

75	**What is the main purpose of chest radiography among PPD-positive contacts?**
A	To detect possible evolution into active TB disease
B	To confirm reinfection from unrelated sources
C	To evaluate environmental causes of dry cough
D	To determine nutritional status and body mass

76	**What public health consequence may occur if MDT supply is interrupted during disasters or conflict?**
A	Increased transmission, complications, and drug resistance
B	Rise in asymptomatic infections only
C	Reduced case detection
D	Spontaneous cure of cases

77	**Which factor determines the need for extended follow-up among contacts?**
A	Financial status, occupation, and education level
B	Cultural background, ethnicity, and travel history
C	Duration of exposure, infectiousness, and immune status
D	Residence size, climate, and dietary preferences

78	**Which programmatic measure supports compliance among high-risk TB contacts?**
A	Exclusive reliance on radiographic monitoring
B	Occasional follow-up only after symptoms appear
C	Directly observed preventive therapy when adherence is uncertain
D	Self-reported adherence without clinical supervision

79	**Which clinical form of leprosy shows strong cell-mediated immunity?**						
A	Indeterminate	B	Borderline	C	Lepromatous	D	Tuberculoid

80	**In a community where tuberculosis incidence is declining, which epidemiological indicator would best demonstrate successful tuberculosis control?**
A	Sustained reduction in incidence of smear-positive tuberculosis
B	Increased number of TB suspects examined
C	Increased BCG vaccination coverage
D	Decreased case fatality rate among treated patients

81	**Which public health strategy is most effective in reducing community transmission of leprosy?**
A	Mass prophylactic antibiotics
B	Routine lepromin testing
C	Early detection and prompt initiation of MDT
D	Isolation of all patients

82	**A public health officer emphasizes health education about the curability of leprosy and availability of free treatment. What major public health problem is this strategy mainly addressing?**
A	Drug toxicity
B	Genetic susceptibility
C	Social stigma and delayed case detection
D	Environmental contamination

83	**Which type of TB is most commonly encountered among reported cases?**
A	Congenital TB transmitted during perinatal period
B	Avian-type TB transmitted from poultry exposure
C	Extra-pulmonary TB including skeletal involvement only
D	Pulmonary TB including smear-positive and smear-negative

84	**What is the recommended duration for standard preventive therapy using isoniazid?**
A	Weekly dosing for immune stimulation only
B	Daily dosing for sufficient long-term protection
C	Quarterly dosing for minimal medication exposure
D	Monthly dosing for stable long-term immunity

85	**Which virus is associated with liver cancer and can be prevented by vaccination?**						
A	HBV	B	HCV	C	HIV	D	HPV

86	**An outbreak investigation identifies a meningococcal carrier rate of 75% among a school population. Which public health action is appropriate?**
A	Avoid prophylaxis and rely on passive environmental surveillance
B	Recommend mass chemoprophylaxis for entire open communities
C	Implement mass chemoprophylaxis only in closed supervised groups
D	Initiate targeted chemoprophylaxis for close contacts only

87	**A 60-year-old man with diabetes and hypertension is advised to reduce salt intake. What is the recommended daily limit?**						
A	1 teaspoon	B	3 teaspoons	C	2 teaspoons	D	No restriction

88	**Which group has the highest risk of rapid progression from TB infection to TB disease?**
A	Travelers without history of previous infections
B	Children older than 12 years with mild symptoms
C	Adults with stable immune systems and low exposure
D	Patients living with HIV due to weakened immunity

89	**Which statement best reflects the global epidemiology of tuberculosis?**
A	One-third of the world population is infected with TB
B	Less than 1% of global TB patients develop complications
C	MDR-TB is not considered a public health emergency
D	TB causes fewer deaths than most acute viral infections

90	**Which organism causes leprosy?**		
A	Mycobacterium leprae	B	Mycobacterium tuberculosis
C	Mycobacterium kansasii	D	Mycobacterium bovis

91	**A contact of a leprosy patient is examined annually for several years even though no symptoms appear. What is the main reason for prolonged follow-up?**
A	Patients develop lifelong immunity immediately
B	The disease has a long and variable incubation period
C	The vaccine requires repeated monitoring
D	The infection spreads through vectors

92	**Which environmental risk factor is LEAST likely to contribute to NCDs?**		
A	Occupational hazards	B	Seasonal influenza
C	Water pollution	D	Air pollution

93	**A national tuberculosis program implements the Directly Observed Treatment Short-course (DOTS) strategy. Which epidemiological outcome is most directly expected?**
A	Increase in latent TB infection
B	Reduction in BCG vaccination coverage
C	Increase in TB transmission
D	Reduction in treatment default and drug resistance

94	**A 30-year-old sedentary office worker develops obesity and hypertension. Which risk factor is most relevant?**		
A	Ethnicity	B	Insufficient physical activity
C	Genetic predisposition	D	Age

95	**Which criterion must be met before beginning preventive therapy for TB contacts?**
A	Confirm latent infection by sputum culture
B	Require hospitalization for baseline assessment
C	Rule out active TB through history and chest X-ray
D	Perform mandatory CT scan for lung mapping

96	**Which WHO treatment regimen is recommended for multibacillary leprosy?**
A	Dapsone alone for 1 year
B	Rifampicin monthly only
C	Rifampicin + Dapsone + Clofazimine for 24 months
D	Rifampicin + Dapsone for 6 months

97	**Which financial burden statement is correct regarding chronic diseases?**
A	Death costs more than living with disease
B	Living with chronic disease is more expensive
C	Medicare spending is minimal on chronic diseases
D	Direct costs always exceed indirect costs

98	**For how long should household contacts of leprosy patients be followed up annually?**						
A	3 years	B	10 years	C	1 year	D	5 years

99	**A patient with lepromatous leprosy develops painful red nodules and systemic symptoms during treatment. Which complication is most likely responsible?**		
A	Reversal reaction	B	Septicemia
C	Acute allergic dermatitis	D	Erythema nodosum leprosum

form 3 ans hic

1. Diagnostic criteria for DM include fasting blood sugar above:

a) 140 mg/dL
b) 125 mg/dL
c) 110 mg/dL
d) 100 mg/dL

Correct Answer: **B**

2. Which principle ensures people contribute financially and with free labor to PHC activities?

a) Multi-sectoral approach
b) Equity
c) Appropriate technology
d) Community participation

Correct Answer: **D**

3. A patient presents 10 days after a Category III dog bite with no prior treatment. What is the BEST management?

a) Full PEP (vaccine + RIG)
b) Observe only
c) No treatment
d) RIG only
e) Vaccine only

Correct Answer: **A**

4. A neonate develops tetanus 1 week after birth at home. Which factor is MOST responsible?

a) Genetic susceptibility
b) Environmental contamination of cord
c) Airborne transmission
d) Person-to-person spread

Correct Answer: **B**

5. Carrier state is MOST important in transmission of:

a) Hepatitis E
b) Hepatitis A
c) Hepatitis B
d) Hepatitis A & E

Correct Answer: **C**

6. Which screening technique is most effective for early detection of cervical cancer?

a) Thermography
b) Endoscopy
c) Pap test (exfoliative cytology)
d) Ultrasound

Correct Answer: **C**

7. Appropriate treatment of common diseases and injuries is part of which PHC element?

a) Promotive services
b) Rehabilitative services
c) Curative services
d) Preventive services

Correct Answer: **C**

8. Which of the following diseases are non-communicable but preventable by vaccination?

a) Cancer & diabetes
b) hepatitis B & C
c) AIDS & typhoid
d) Tetanus & Rabies

Correct Answer: **D**

9. Which of the following findings most strongly indicates failure of the surveillance system in rabies control programs?

a) Timely reporting of suspected animal bite cases
b) Implementation of public health education activities
c) Underreporting of animal bite exposures
d) High coverage of mass dog vaccination campaigns

Correct Answer: **C**

10. The long incubation period of rabies affects control by:

a) Making diagnosis easy
b) Preventing vaccination
c) Allowing delayed intervention
d) Increasing immediate mortality

Correct Answer: **C**

11. Which level of prevention is wound management?

a) Secondary
b) Primordial
c) Tertiary
d) Primary

Correct Answer: **D**

12. A fully immunized adult sustains a clean surgical wound. Last booster was 12 years ago. What is the MOST appropriate management?

a) TIG only
b) TT booster only
c) TT + TIG
d) No action

Correct Answer: **B**

13. Tetanus prevention mainly depends on:

a) Isolation
b) Vaccination
c) Antibiotics
d) Quarantine

Correct Answer: **B**

14. A 52 year old man with hypertension and obesity develops CAD. Which of the following is a non-modifiable risk factor?

a) Age
b) Hypertension
c) Smoking
d) Obesity

Correct Answer: **A**

15. What conditions can chronic infection with Helicobacter pylori predispose to?

a) Lung cancer
b) Colon cancer
c) Gastric cancer
d) Liver cancer

Correct Answer: **C**

16. Which preventive measure is common to both CAD and DM?

a) Regular physical activity and weight control
b) Avoidance of viral infections
c) Genetic counseling for HLA linkage
d) Avoidance of oral contraceptives

Correct Answer: **A**

17. Which hepatitis is MOST associated with fecal-oral transmission?

a) D
b) B
c) A
d) C

Correct Answer: **C**

18. In the context of hepatitis B virus (HBV) infection, which of the following represents indirect transmission of the pathogen?

a) Direct skin-to-skin contact with an infected person
b) Exposure to respiratory droplets from a coughing patient
c) Inhalation of virus-laden airborne particles
d) Receiving contaminated blood via transfusion

Correct Answer: **D**

19. Alcohol consumption is most strongly linked to which cancer?

a) Skin cancer
b) Pancreatic cancer
c) Oral cancer
d) Kidney cancer

Correct Answer: **C**

20. For hepatitis B, which of the following community-level interventions is the most effective for reducing disease incidence?

a) Isolation of symptomatic individuals
b) Encouraging bed rest for infected individuals
c) Vaccination + safe blood and injection practices
d) Empirical use of antiviral medications

Correct Answer: **C**

21. A newborn develops tetanus. The MOST likely cause is:

a) Genetic defect
b) Breastfeeding
c) Airborne infection
d) Poor umbilical hygiene

Correct Answer: **D**

22. Communication barrier includes negative attitudes towards the message. Which of the following reflects this barrier?

a) Enthusiastic acceptance
b) Interest in the topic
c) Receptiveness
d) Resistance to communication

Correct Answer: **D**

23. A 55-year-old obese woman develops endometrial cancer. Which risk factor is most relevant?

a) Genetic predisposition
b) Radiation exposure
c) Obesity and physical inactivity
d) Viral infection

Correct Answer: **C**

24. Which level of prevention of hepatitis C is screening blood donors?

a) Tertiary
b) Primary
c) Primordial
d) Secondary

Correct Answer: **B**

25. Which of the following is considered the most sensitive public health indicator for assessing the effectiveness of tetanus prevention and maternal immunization programs?

a) Tetanus-related mortality in hospitalized patients
b) Number of reported wound infections
c) Incidence of neonatal tetanus
d) Overall incidence of tetanus in adults

Correct Answer: **C**

26. A 25-year-old thin male presents with polyuria, polydipsia, and sudden weight loss. Which type of diabetes is most likely?

a) Type 2 DM
b) Secondary DM due to drugs
c) MODY
d) Type 1 DM

Correct Answer: **D**

27. What health risks does a farmer face when exposed to aflatoxin from contaminated maize?

a) Pancreatic cancer
b) Hepatocellular carcinoma
c) Gastric cancer
d) Lung cancer

Correct Answer: **B**

28. In Primary Health Care, oral rehydration therapy (ORT) for diarrheal disease control is an example of what principle?

a) Multi-sectoral approach
b) Community participation
c) Equity
d) Appropriate technology

Correct Answer: **D**

29. Viral infections such as rubella and mumps are associated with which type of DM?

a) Type 1 DM
b) MODY
c) Secondary DM due to drugs
d) Type 2 DM

Correct Answer: **A**

30. Involving people in planning and evaluation of PHC services is an example of which PHC principle?

a) Appropriate technology
b) Equity
c) Community participation
d) Multi-sectoral approach

Correct Answer: **C**

31. Which physical agent is a major carcinogen?

a) Aflatoxin
b) Ultraviolet radiation
c) Benzene
d) HPV

Correct Answer: **B**

32. A veterinarian is scratched by a vaccinated dog that can be observed. What is the BEST approach?

a) Vaccine only
b) Observe animal without starting PEP
c) Immediate full PEP with RIG
d) Start vaccine, discontinue if dog remains healthy

Correct Answer: **D**

33. A farmer presents with a contaminated wound. His immunization status is unknown, What is the BEST management?

a) TT only
b) TT + TIG
c) TIG only
d) Antibiotics only

Correct Answer: **B**

34. Which group is MOST at risk for rabies exposure?

a) Teachers
b) Retail workers
c) Veterinarians
d) Office workers

Correct Answer: **C**

35. Which group is HIGH risk for hepatitis B?

a) Teachers
b) Farmers
c) Drivers
d) Healthcare workers

Correct Answer: **D**

36. A child presents with a deep contaminated wound caused by a fall and a simultaneous bite from a stray dog. Immunization history is unknown. What is the MOST appropriate management?

a) TT + rabies vaccine
b) Rabies vaccine only
c) TT only
d) TT + TIG + rabies vaccine + RIG

Correct Answer: **D**

37. Which strategy reflects primary prevention of rabies?

a) Dog population control
b) Early diagnosis
c) Rehabilitation
d) Immunoglobulin after bite

Correct Answer: **A**

38. A limitation of mass communication is that it does what?

a) Gives knowledge without guaranteed behavior change
b) Influences attitudes deeply
c) Provides direct feedback
d) Is suitable for confidential issues

Correct Answer: **A**

39. A 6-year-old child presents after a deep bite from a stray dog that escaped. The child completed full rabies vaccination 2 years ago. What is the MOST appropriate management?

a) No treatment
b) Wound care + rabies vaccine (no RIG)
c) Rabies immunoglobulin (RIG) only
d) Rabies vaccine only

Correct Answer: **B**

40. Vertical transmission is MOST common in:

a) Hepatitis B
b) Hepatitis E
c) Hepatitis D
d) Hepatitis A

Correct Answer: **A**

41. Which is the MOST critical step in rabies post-exposure management?

a) Immediate wound washing
b) Isolation
c) Blood transfusion
d) Antibiotics

Correct Answer: **A**

42. Which scenario best fits "health appraisal" in individual education?

a) Community gathering
b) Comprehensive medical exam with screening tests
c) School lecture
d) Mass media awareness

Correct Answer: **B**

43. A community health worker is planning an awareness campaign about diabetes. Which strategy is most effective for long-term behavior change?

a) Providing free glucose meters
b) Broadcasting radio advertisements
c) Conducting interactive workshops
d) Distributing pamphlets only

Correct Answer: **C**

44. Which of the following is a modifiable risk factor for both CAD and DM?

a) Age
b) Family history
c) Obesity
d) Male sex

Correct Answer: **C**

45. A major disadvantage of the personal approach is that it reaches only a limited number of people. Why is this a limitation?

a) It cannot reach mass audiences
b) It becomes too expensive
c) It requires too many posters
d) It eliminates communication barriers

Correct Answer: **A**

46. Which is the MAIN reservoir of tetanus?

a) Water
b) Animals
c) Humans
d) Soil

Correct Answer: **D**

47. Familial clustering is most strongly associated with which cancer?

a) Cervical cancer
b) Gastric cancer
c) Colon cancer
d) Liver cancer

Correct Answer: **C**

48. Rabies is classified epidemiologically as:

a) Zoonotic disease
b) Nosocomial infection
c) Food-borne disease
d) Vector-borne disease

Correct Answer: **A**

49. In a rabies-endemic rural area, which intervention will have the GREATEST population impact?

a) Human PEP only
b) Pre-exposure vaccination for all children
c) Mass dog vaccination
d) Isolation of patients

Correct Answer: **C**

50. Which function of communication focuses on influencing the receiver to adopt a behavior as part of regular life?

a) Directing behavior change
b) Entertainment
c) Reinforcing knowledge
d) Establishing social relationships

Correct Answer: **A**

51. The role of municipalities in sanitation illustrates which PHC principle?

a) Community participation
b) Multi-sectoral approach
c) Appropriate technology
d) Equity

Correct Answer: **B**

52. Which factor accounts for about 30% of cancers?

a) Radiation
b) Diet and nutrition
c) Genetic susceptibility
d) Occupational exposure

Correct Answer: **B**

53. Communication in health education includes components such as communicator, message, and recipients. Who is considered the communicator?

a) Originator of the message
b) Environment
c) Patient
d) Recipient

Correct Answer: **A**

54. A patient develops tetanus following a contaminated wound with no prior prophylaxis. Which level of prevention failed?

a) Primary
b) Tertiary
c) Primordial
d) Secondary

Correct Answer: **A**

55. A 70-year-old man develops prostate cancer. Which statement about sex distribution of cancers is correct?

a) Men only develop reproductive cancers
b) Men and women have equal incidence
c) Men have higher overall cancer incidence
d) Women have higher overall cancer incidence

Correct Answer: **C**

56. Booster doses of tetanus toxoid are important because:

a) Vaccine is ineffective
b) Transmission increases
c) Immunity wanes over time
d) Disease mutates

Correct Answer: **C**

57. Which intervention reduces tetanus mortality MOST?

a) Routine immunization
b) Isolation
c) Early ICU care
d) Antibiotics

Correct Answer: **A**

58. Which intervention prevents hepatitis B transmission?

a) Isolation
b) Mosquito control
c) Boiling water
d) Vaccination

Correct Answer: **D**

59. In a low-resource rural setting with high incidence of both rabies and tetanus, which strategy is MOST effective?

a) Case management only
b) Improve ICU care
c) Antibiotic distribution
d) Mass dog vaccination + human immunization + health education

Correct Answer: **D**

60. A rise in hepatitis A cases suggests failure of:

a) Screening
b) Vector control
c) Immunization
d) Sanitation

Correct Answer: **D**

61. A community reports increased dog bites. Which measure is MOST effective in reducing rabies incidence at population level?

a) Post-exposure vaccination only
b) Isolation of bite victims
c) Mass dog vaccination
d) Antibiotic prophylaxis

Correct Answer: **C**

62. Which dietary pattern is protective against type 2 DM?

a) High sugar diet
b) High saturated fat diet
c) Prudent diet rich in fruits and vegetables
d) Conservative diet rich in butter and potatoes

Correct Answer: **C**

63. What health risks are slightly increased by oral contraceptive use?

a) Breast cancer
b) Cervical cancer
c) Ovarian cancer
d) Endometrial cancer

Correct Answer: **A**

64. Panel discussions involve a group of 4–8 people discussing a topic in front of an audience. What is this an example of?

a) Environmental approach
b) Individual education
c) Mass communication
d) Group health education

Correct Answer: **D**

65. For dietary prevention of CAD, saturated fats should constitute less than what percentage of total energy intake?

a) 15%
b) 10%
c) 5%
d) 20%

Correct Answer: **B**

66. Which intervention is MOST cost-effective in Rabies endemic regions?

a) Isolation units
b) Human vaccination only
c) ICU care
d) Animal vaccination programs

Correct Answer: **D**

67. Neonatal tetanus indicates failure of:

a) Maternal immunization
b) Vector control
c) Secondary prevention
d) Surveillance

Correct Answer: **A**

68. A 60-year-old hypertensive patient is advised lifestyle modification. Which measure is most effective in reducing CAD incidence?

a) Increased caffeine intake
b) Low salt diet
c) High protein diet
d) Vitamin supplementation

Correct Answer: **B**

69. Which group is MOST vulnerable for Tetanus?

a) Elderly fully immunized
b) Unimmunized individuals
c) Healthcare workers
d) Vaccinated adults

Correct Answer: **B**

70. Health campaigns such as road safety demonstrations rely on what?

a) Silent posters only
b) Individual interviews
c) Advanced therapies
d) Shock tactics

Correct Answer: **D**

71. In Primary Health Care (PHC), what does equity mean?

a) Services for urban areas only
b) Services for all, more for vulnerable
c) Services based on ability to pay
d) Services only for the rich

Correct Answer: **B**

72. A farmer bitten by a stray dog delays seeking care. Which level of prevention failed?

a) Primary
b) Primordial
c) Tertiary
d) Secondary

Correct Answer: **A**

73. A 48 year old woman on oral contraceptives develops CAD. Which category of risk factor does this belong to?

a) Environmental
b) Modifiable
c) Genetic
d) Non-modifiable

Correct Answer: **B**

74. Communication serves several functions in health education such as establishing relationships and delivering new information. Which of the following is a function of communication?

a) Reinforcing knowledge
b) Encouraging silence
c) Destroying relationships
d) Providing misinformation

Correct Answer: **A**

75. Which complication of diabetes increases risk of ischemic heart disease 3.5–4 times higher?

a) Neuropathy
b) Retinopathy
c) Foot ulcers
d) Macrovascular atherosclerosis

Correct Answer: **D**

76. Which principle ensures services reach vulnerable groups through outreach programs?

a) Appropriate technology
b) Community participation
c) Multi-sectoral approach
d) Equity

Correct Answer: **D**

77. Health education promotes disease prevention. Which prevention level does health education mainly represent?

a) Secondary prevention
b) Curative care
c) Tertiary prevention
d) Primary prevention

Correct Answer: **D**

78. Chronic infection is MOST associated with:

a) Hepatitis B
b) Hepatitis A
c) Hepatitis E
d) Food poisoning

Correct Answer: **A**

79. Which factor MOST contributes to rabies persistence in endemic areas?

a) Human-to-human transmission
b) Animal reservoir maintenance
c) Airborne spread
d) Short incubation period

Correct Answer: **B**

80. Promotion of food supply and proper nutrition is classified under which PHC element?

a) Preventive
b) Promotive
c) Curative
d) Rehabilitative

Correct Answer: **B**

formative 4 hic ans

1. A 45-year-old man living near a busy industrial zone reports increasing respiratory irritation. The environment around him contains multiple physical, chemical, and biological hazards. According to environmental health principles, which factor best describes the external influences affecting his health?

a) Internal metabolic changes
b) Environmental conditions surrounding him
c) Genetic predisposition
d) Personal lifestyle only

Correct Answer: **B**

2. A child in a rural home repeatedly develops diarrhea due to contaminated water. What is the most critical housing criterion missing?

a) Adequate ventilation
b) Indoor temperature regulation
c) Structural safety
d) Safe water supply

Correct Answer: **D**

3. A chlorination unit aims to maintain a residual chlorine level after 30 minutes. What is the ideal residual range?

a) 5–10 ppm
b) 1–1.2 ppm
c) 0 ppm
d) 0.1–0.2 ppm

Correct Answer: **D**

4. A city experiencing winter smog finds sunlight insufficient to break down pollutants. Which self-cleansing mechanism is failing?

a) Plant life
b) Sunlight
c) Rain
d) Wind

Correct Answer: **B**

5. A community experiences fog mixed with pollutants resulting in high mortality, similar to the 1952 London smog. What type of pollution effect is this?

a) Global effect

b) Long-term effect

c) Immediate health effect

d) Psychological effect

Correct Answer: **C**

6. A community near a desert suffers from frequent dust storms. Which natural factor normally helps cleanse air but is overloaded here?

a) Rain

b) Sunlight

c) Wind

d) Plant life

Correct Answer: **C**

7. A community reports an increase in asthma cases following the establishment of a large textile factory. Which environmental component is directly implicated in these health effects?

a) Cultural environment

b) Physical environment

c) Social environment

d) Biological environment

Correct Answer: **B**

8. A disabled elderly man cannot safely enter his bathroom due to narrow doorways. Which criterion is lacking?

a) Accessibility

b) Water supply

c) Space adequacy

d) Vector protection

Correct Answer: **A**

9. A factory worker is exposed to high noise levels and heat, causing stress and fatigue. These risks fall under which hazard category?

a) Cultural

b) Physical

c) Social

d) Biological

Correct Answer: **B**

10. A farmer using a surface water canal notices contamination from upstream sewage discharge. Which water source type is this?

a) Rainwater
b) Surface water
c) Groundwater
d) Distilled seawater

Correct Answer: **B**

11. A filtration plant reports decreasing flow rate due to a thickened biological layer. What is this layer known?

a) Sedimentation sludge
b) Surface film
c) Dorr clarifier
d) Chlorine residue

Correct Answer: **B**

12. A low-income neighborhood suffers from overcrowded homes and limited access to healthcare services. Which environmental component is contributing to their poor health?

a) Biological
b) Social
c) Occupational
d) Cultural

Correct Answer: **B**

13. A man develops chronic bronchitis after long-term exposure to poor air quality. This represents which air pollution effect?

a) Psychological effect
b) Immediate effect
c) Global effect
d) Long-term effect

Correct Answer: **D**

14. A municipal water sample shows albuminoid ammonia of 0.5 ppm. What this indicates?

a) Balanced pH
b) Excessive organic matter
c) Pure groundwater
d) Acceptable levels

Correct Answer: **B**

15. A newly built home near a forest fails to protect occupants from mosquitoes. Which criterion is missing?

a) Good ventilation
b) Chemical-free materials
c) Vector protection
d) Adequate temperature control

Correct Answer: **C**

16. A nuclear plant accident releases harmful radiation. Which global pollution type does this represent?

a) Ozone depletion
b) Greenhouse effect
c) Acid rain
d) Radioactive disaster

Correct Answer: **D**

17. A person complains of bitterness in well water due to high bicarbonates. This is an example of what?

a) Permanent hardness
b) Temporary hardness
c) Nitrite contamination
d) Organic pollution

Correct Answer: **B**

18. A polluted city uses trees extensively in urban design to improve air quality. Which natural mechanism are they maximizing?

a) Oxygen generation
b) Evaporation
c) Bacterial growth
d) Gas oxidation

Correct Answer: **A**

19. A population exhibits unhealthy food habits and cultural practices that increase exposure to unhygienic environments. Which environmental component contributes most to these outcomes?

a) Biological
b) Industrial
c) Physical
d) Cultural

Correct Answer: **D**

20. A public health officer aims to prevent disease by controlling environmental hazards in communities. This objective aligns with which WHO-defined field?

a) Environmental sanitation
b) Nutritional health
c) Genetic medicine
d) Clinical epidemiology

Correct Answer: **A**

21. A rapid sand filter reduces bacteria to approximately what proportion of original load?

a) 100%
b) 90%
c) 50%
d) 5%

Correct Answer: **D**

22. A rural community suffers from rodent infestation, unsafe food handling, and unclean homes. Improving these factors directly supports which objective of environmental sanitation?

a) Preventing disease via hygiene
b) Increasing industrial productivity
c) Improving genetic quality
d) Reducing radiation exposure

Correct Answer: **A**

23. A rural house built from unstable materials collapses during rainfall. Which housing standard was violated?

a) Accessibility
b) Structural safety
c) Ventilation
d) Temperature regulation

Correct Answer: **B**

24. A sample shows oxygen absorption of 6 ppm. What this indicates?

a) Heavy organic pollution
b) Mineral deficiency
c) Over softened water
d) Safe water

Correct Answer: **A**

25. A survey reveals that residents near a swampy area have increased exposure to mosquitoes, parasites, and pathogens. Which environmental component primarily affects their health?

a) Social
b) Physical
c) Biological
d) Cultural

Correct Answer: **C**

26. A technician tests water and finds no coliforms in 100 ml. This meets which drinking-water standard?

a) Biological
b) Radioactive
c) Chemical
d) Physical

Correct Answer: **A**

27. A town decides to build water intake pipes upstream from the city to reduce contamination. What does this structural approach ensure?

a) Better chlorination
b) Better pH control
c) Avoidance of pollution sources
d) Increased sedimentation time

Correct Answer: **C**

28. A village near a volcano experiences ash fallout that reduces air quality. This is an example of which pollution source?

a) Domestic
b) Industrial
c) Vehicular
d) Natural

Correct Answer: **D**

29. A woman living near a busy road develops worsening asthma. Which pollutant is most associated with vehicle emissions?

a) Asbestos
b) Hydrogen cyanide
c) Aluminum hydroxide
d) Carbon monoxide

Correct Answer: **D**

30. A woman reports her drinking water tastes unpleasant and appears cloudy. Which property of safe drinking water is violated?

a) Radioactive free

b) Free of pathogens

c) Free of chemicals

d) Clear and pleasant

Correct Answer: **D**

31. An area experiences increased UV radiation due to atmospheric changes. Which global air pollution effect explains this?

a) Dioxin pollution

b) Ozone layer depletion

c) Greenhouse effect

d) Acid rain

Correct Answer: **B**

32. An increase in particulate matter is detected in winter due to stagnant air. Which factor influences these levels?

a) Vegetation density

b) Air temperature and humidity

c) Soil alkalinity

d) Water pH

Correct Answer: **B**

33. Burning plastics generates toxins that cause immune suppression and reproductive damage. What these substances are called?

a) Asphyxiants

b) Silicates

c) Nitrites

d) Dioxins

Correct Answer: **D**

34. Damp, poorly ventilated housing increases the risk of which of the following?

a) Radiation sickness

b) Skin cancers

c) Malaria

d) Respiratory diseases

Correct Answer: **D**

35. Improper disposal of sewage into rivers violates which water pollution prevention measure?

a) Waste management
b) Industrial regulation
c) Agricultural practices
d) Global cooperation

Correct Answer: **A**

36. Overcrowded homes with poor privacy most affect which health domain?

a) Occupational health
b) Radiological health
c) Genetic health
d) Mental health

Correct Answer: **D**

37. Presence of E. coli in water indicates which of the following?

a) No contamination
b) Industrial pollution
c) Remote contamination
d) Recent fecal contamination

Correct Answer: **D**

38. Smoke from households burning firewood contributes mostly to which pollution source category?

a) Miscellaneous
b) Domestic
c) Industrial
d) Natural

Correct Answer: **B**

39. Switching from fossil fuels to solar and wind energy addresses which pollution control approach?

a) Industrial filtration
b) Transport intervention
c) Household measures
d) Energy transition

Correct Answer: **D**

40. Which bacterium indicates remote fecal contamination due to long survival?

a) E. coli
b) Clostridium welchii
c) Klebsiella aerogenes
d) Streptococcus faecalis

Correct Answer: **B**

41. Which compound indicates recent organic pollution when present in drinking water?

a) Nitrates
b) Albuminoid ammonia
c) Carbonate hardness
d) Nitrites

Correct Answer: **D**

42. Which water source is most prone to collecting atmospheric dust and bacteria?

a) Deep wells
b) Rivers
c) Rainwater
d) Springs

Correct Answer: **C**

Formative 5

1. No HCV vaccine means prevention relies on?

(a) Harm-reduction strategies
(b) Breastfeeding avoidance
(c) Handwashing only
(d) Cooking food

2. Primary HCV prevention?

(a) Air filters
(b) Safe injections
(c) Cough hygiene
(d) Masks

3. A pottery worker develops fibrosis after inhaling dust containing free crystalline silica. Which mechanism best explains the pathology?

(a) Direct allergic reaction
(b) Rupture of macrophages releasing fibrogenic substances
(c) Excessive surfactant secretion
(d) Activation of eosinophils

4. In HIV context, Mother-to-child transmission reduced from 35% to 1% due to?

(a) Triple ART in pregnancy
(b) Cesarean sections
(c) No breastfeeding
(d) More midwives

5. A textile worker develops asthma and recurrent urticaria after exposure to certain workplace dyes. What category is responsible?

(a) Fumes
(b) Allergens
(c) Irritants
(d) Asphyxiants

6. HCV diagnosis confirmed by?

(a) X-ray
(b) Glucose test
(c) Anti-HCV only
(d) RNA testing

7. A 27-year-old man presents with fever and lymphadenopathy 3 weeks after unprotected intercourse; rapid HIV test is negative. What is the next best step?

(a) Perform CD4 count
(b) Perform HIV RNA viral load test
(c) Repeat ELISA after 1 year
(d) Start ART immediately without testing

8. Chronic farmer's lung may lead to which of the following?

(a) Dermatitis
(b) Liver cirrhosis
(c) Cor pulmonale
(d) Pleural plaques

9. HBV stable on surfaces for?

(a) 7 days
(b) 1 hour
(c) 1 year
(d) 30 days

10. Leading ensures?

(a) Silence
(b) Isolation
(c) Avoiding communication
(d) Motivation and teamwork

11. A welder with no eye protection develops severe eye pain, photophobia, and lacrimation after work. What is the most likely diagnosis?

(a) Welder's flash
(b) Corneal ulcer
(c) Retinal burn
(d) Acute glaucoma

12. A farmer presents with recurrent episodes of fever, dyspnea, and fine nodular densities on X ray in winter. Most likely diagnosis?

(a) Asthma
(b) Bagassosis
(c) Farmer's lung
(d) Silicosis

13. Birth spacing prevents?

(a) Childhood growth

(b) Eating disorders

(c) Maternal depletion

(d) School attendance

14. Poor intranatal asepsis leads to?

(a) Less labor pain

(b) Increased satisfaction

(c) Infection risks

(d) Better neonatal outcomes

15. Best HBV control in blood banks?

(a) Donor color check

(b) Smelling blood

(c) Screening all donations

(d) No tests

16. A pregnant woman newly diagnosed HIV positive. What predicts transmission risk?

(a) Maternal BMI

(b) Maternal viral load

(c) Breastfeeding status

(d) Mode of delivery

17. Tattooing risk when?

(a) Wearing gloves

(b) Using sterile disposables

(c) Using unsterile instruments

(d) Using ink only

18. What are earplugs used for?

(a) Radiation

(b) Noise exposure

(c) Vibration

(d) Chemical fumes

19. A TB control program under DOTS achieves high cure rates but experiences drug wastage and redundant testing. This indicates:

(a) High effectiveness but low efficiency

(b) High efficiency and high effectiveness

(c) Neither efficiency nor effectiveness

(d) High efficiency but low effectiveness

20. Delegation includes?

(a) No communication

(b) Avoiding responsibility

(c) Granting authority and accountability

(d) Doing everything alone

21. To reduce stigma blocking HIV testing, best approach?

(a) Mandatory disclosure

(b) HIV quarantine

(c) Confidential community-based testing

(d) Travel bans

22. Best strategy for reducing HIV in sex workers?

(a) ART only

(b) Isolation

(c) Imprison clients

(d) Condoms + testing + empowerment

23. What do occupational physician responsibilities include?

(a) First aid, exams, health education

(b) Plant construction

(c) Salary administration

(d) Electrical repairs

24. What is the most important preventive measure for pneumoconiosis?

(a) Steroids

(b) Antibiotics

(c) Dust control

(d) Bronchodilators

25. HIV+ mother in safe formula environment should?

(a) Exclusively breastfeed

(b) Breastfeed at night only

(c) Stop ART

(d) Avoid breastfeeding

26. What is the most dangerous particle size for alveolar penetration?

(a) 0.5–3 μm

(b) 10–5 μm

(c) <0.1 μm

(d) >10 μm

27. High-risk pregnancies require?

(a) Early referral and monitoring

(b) Home delivery

(c) No referral

(d) Delayed care

28. Health education on danger signs helps?

(a) No impact

(b) Early care-seeking

(c) Delayed care

(d) Avoiding clinics

29. Emergency obstetric care includes?

(a) C-sections and management of complications

(b) No interventions

(c) Dance classes

(d) Music therapy

30. What does segregation in industrial safety mean?

(a) Grouping workers

(b) Separating hazardous processes

(c) Separating offices

(d) Increasing production lines

31. Postpartum danger requiring urgent action?

(a) Heavy postpartum hemorrhage

(b) Light bleeding

(c) Mild swelling

(d) Appetite increase

32. Unsafe injections increase?

(a) Diabetes

(b) Malaria

(c) HBV/HCV transmission

(d) Pneumonia

33. Chronic HBV carrier indicates?

(a) Immunity

(b) No infection

(c) Immediate recovery

(d) HBsAg positive >6 months

34. A textile worker develops breathlessness specifically on the first day after return from weekend vacation. What is the likely diagnosis?

(a) Farmer's lung

(b) Byssinosis

(c) Silicosis

(d) Bagassosis

35. A worker with exposure to cotton dust but no moulds is unlikely to develop which of the following?

(a) Farmer's lung

(b) Weaver's cough

(c) Mill fever

(d) Byssinosis

36. Maternal mortality high due to?

(a) Poor access to emergency obstetric care

(b) Clean water

(c) High literacy

(d) Exercise

37. A worker exposed to silica dust for 12 years presents with dyspnea and nodular opacities in upper lobes. Which complication is MOST expected?

(a) Hepatitis

(b) Glaucoma

(c) Dermatitis

(d) Tuberculosis

38. Planning involves?

(a) Reacting only

(b) Ending operations

(c) Setting goals in advance

(d) Waiting

39. Organizational structure determined by?

(a) Functions and duties

(b) Random hierarchy

(c) Staff mood

(d) Luck

40. Increase in transfusion-transmitted HIV. Best prevention?

(a) CBC screening

(b) Stop blood donations

(c) Screen only symptomatic donors

(d) Universal blood screening

41. Occupational dermatitis is caused mainly by which of the following?

(a) Noise

(b) Irritants

(c) Vibration

(d) Dust

42. Multiple previous C-sections categorizes pregnancy as?

(a) Normal

(b) No risk

(c) Low-risk

(d) High-risk

43. HBIG given to newborn?

(a) At 1 year

(b) Never

(c) Within 12 hours of birth

(d) During labor

44. Key objective of maternal programs?

(a) Increase C-sections

(b) Reduce maternal mortality

(c) Eliminate counseling

(d) Increase hospital profits

45. A 45 year old furnace worker collapses after prolonged exposure to high radiant heat. Examination shows normal electrolytes, but marked peripheral pooling of blood and transient loss of consciousness. What is the most likely diagnosis?

(a) Heat syncope

(b) Heat exhaustion

(c) Heat cramps

(d) Heat stroke

46. HBV chronicity highest in?

(a) Elderly only

(b) Adults

(c) Infants

(d) Teens only

47. Hepatitis B transmission risk highest in?

(a) Tears

(b) Saliva only

(c) Blood and body fluids

(d) Sweat

48. HCV persists due to?

(a) No cells infected

(b) Viral RNA persistence

(c) DNA repair

(d) No replication

49. Healthy-feeling HIV patient declines ART; explain?

(a) ART weakens immunity

(b) Early ART prevents onward transmission

(c) ART unnecessary if asymptomatic

(d) ART is only for advanced cases

50. New HIV strain increases infectivity due to?

(a) Agent variation

(b) Climate change

(c) Host resistance

(d) Poverty only

51. Controlling includes?

(a) Punishment only

(b) No feedback

(c) Ignoring deviations

(d) Measuring performance

52. What does social worker in occupational health focus on?

(a) Improving interpersonal relations

(b) Radiation measurement

(c) Collecting taxes

(d) Managing machines

53. A mutation causing faster CD4 decline indicates change in?

(a) Environment

(b) Host factors

(c) Behavior only

(d) Agent factors

54. Asymptomatic HIV with CD4 650 indicates?

(a) Non-transmissible stage

(b) Requires hospitalization

(c) AIDS

(d) Chronic HIV infection

55. Management by exception focuses on?

(a) No priorities

(b) Random noise

(c) Critical deviations

(d) All routine details

56. A worker exposed to asbestos presents with pleural plaques and shaggy cardiac borders on X ray. Which is the pathognomonic finding?

(a) Honeycomb lung

(b) Cavitations

(c) Pleural plaques

(d) Hilar lymphadenopathy

57. A new cotton worker develops fever, chills, and cough within hours of exposure. Which condition is likely?

(a) Farmer's lung

(b) Weaver's cough

(c) Mill fever

(d) Byssinosis

58. Child development depends on?

(a) Music

(b) Maternal health

(c) Fashion

(d) Weather

59. Key step in secondary HIV prevention?

(a) ART for AIDS only

(b) Condom distribution

(c) Early diagnosis and partner notification

(d) Needle reuse training

60. What does substitution in occupational health refer to?

(a) Replacing hazardous materials

(b) Replacing workers

(c) Replacing equipment

(d) Replacing shift patterns

61. Coordination ensures?

(a) Random effort

(b) Isolation

(c) Conflict creation

(d) Alignment toward common goals

62. A woman presents late for antenatal care. Which component improves outcomes?

(a) Screening for high-risk conditions

(b) Avoiding supplements

(c) No counseling

(d) Reducing clinic visits

63. Danger sign requiring immediate care?

(a) Light fatigue

(b) Hunger

(c) Mild nausea

(d) Swelling of feet with headache

64. Division of labor improves organizational performance mainly by:

(a) Reducing worker responsibility

(b) Eliminating the need for supervision

(c) Enhancing efficiency and specialization

65. HCV can lead to?

(a) Cirrhosis and liver cancer

(b) Diabetes only

(c) Thyroid storm

(d) Skin cancer

66. Why screen all pregnant women for HIV?

(a) Research

(b) Prevent mother-to-child transmission

(c) Increase surgical deliveries

(d) Ban breastfeeding

67. Perinatal HBV transmission highest when mother is?

(a) Anti-HBs positive

(b) HBeAg positive

(c) HBcAg negative

(d) HBcIgM positive only

68. HBV vaccine protects for?

(a) 1 year

(b) Lifetime guaranteed

(c) 15+ years

(d) 2 weeks

69. In management, SMART goals require?

(a) No timeframe

(b) Unlimited goals

(c) Random targets

(d) Specific measurable objectives

70. Hemodialysis patients at high risk for?

(a) HBV

(b) Anemia only

(c) Acne

(d) Flu

71. Antenatal care reduces mortality by?

(a) Increasing labor duration

(b) Early detection of complications

(c) Providing surgeries

(d) Reducing breastfeeding

72. Good leadership requires?

(a) Communication and motivation

(b) Favoritism

(c) No feedback

(d) Silence

73. Injection drug users drive rising HIV transmission. Best intervention?

(a) Universal HIV isolation

(b) Mass media campaigns

(c) Reducing school hours

(d) Needle and syringe exchange programs

74. A worker in a glass manufacturing plant presents with confusion, delirium, hot dry skin, and a rectal temperature of 41°C. Which condition most likely explains his symptoms?

(a) Heat exhaustion

(b) Heat hyperpyrexia

(c) Heat stroke

(d) Heat syncope

75. Multiple pregnancy risk arises from?

(a) Air pollution

(b) High vitamin intake

(c) Increased obstetric complications

(d) Hair treatment

76. The decision cycle ends with?

(a) Endless meetings

(b) Rewriting goals only

(c) Executing the best option

(d) Ignoring results

77. What does periodic medical exam primarily aim to?

(a) Detect early reversible disease

(b) Improve productivity

(c) Reduce taxes

(d) Promote company profit

78. Photophobia and keratitis from snow reflection is due to which of the following?

(a) UV

(b) IR

(c) Microwaves

(d) Gamma rays

79. A worker exposed to silica dust shows accelerated progression of lung fibrosis within only 2 years. Which factor most likely influenced this?

(a) Good nutrition

(b) Low dust concentration

(c) Cold exposure

(d) Very small dust size

80. Post-exposure HBV prophylaxis?

(a) HBIG + vaccine

(b) Nothing

(c) Water only

(d) Soap

81. In a region with poor adherence to antiretroviral therapy (ART), which of the following indicators is the most sensitive and specific for monitoring program effectiveness?

(a) Viral load suppression rate

(b) ART retention rate at 12 months

(c) HIV incidence rate

(d) CD4 cell count improvement

82. First antenatal visit goal?

(a) Avoid all tests

(b) Confirm pregnancy and detect risks early

(c) Schedule delivery immediately

(d) Weight loss plan

83. Poor maternal nutrition affects fetus via?

(a) Adult obesity only

(b) No effect

(c) Skin changes

(d) Increased perinatal risks

84. CD4 150 in HIV patient predisposes to?

(a) Hepatitis A

(b) Tetanus

(c) Influenza

(d) Pneumocystis pneumonia

85. What is the best time for pre placement exam?

(a) After hiring

(b) During illness

(c) Before employment

(d) At retirement

86. Postnatal follow-up aims to?

(a) Delay vaccines

(b) Stop breastfeeding

(c) Detect complications early

(d) Reduce bonding

87. In a region with rising HIV among youth, which preventive measure has strongest epidemiological impact?

(a) Increasing ART clinics

(b) Comprehensive sexual education

(c) Promoting abstinence only

(d) Increasing ICU beds

88. A machine operator experiences marked irritability, hypertension, and persistent tinnitus after working near loud machinery for years. Which noise effect predominates?

(a) Non auditory—physiological

(b) Auditory—cochlear damage

(c) Mechanical trauma

(d) Psychological stress only

89. Healthcare workers need?

(a) Diet plan

(b) No vaccination

(c) HBV vaccination

(d) Yoga

90. Management improves productivity by?

(a) No delegation

(b) Optimal resource utilization

(c) Chaos

(d) Avoiding structure

91. A patient with HIV has CD4 480 and chronic diarrhea. Interpretation?

(a) Terminal stage

(b) AIDS defining condition

(c) Moderate immunosuppression

(d) Not HIV-related

92. Decision-making requires?

(a) Guessing

(b) Luck

(c) Delays

(d) Accurate information

93. Which of the following best differentiates efficiency from effectiveness in health systems?

(a) Efficiency and effectiveness are interchangeable terms

(b) Efficiency focuses on optimal resource use, while effectiveness focuses on achieving intended health outcomes

(c) Efficiency focuses on achieving desired health outcomes, while effectiveness focuses on cost reduction

(d) Efficiency focuses on service coverage, while effectiveness focuses on staffing

94. Skilled birth attendance prevents?

(a) Jaundice

(b) Postpartum depression

(c) Delivery complications

(d) Childhood obesity

95. Organizing includes?

(a) No authority

(b) Random tasking

(c) Grouping activities logically

(d) No roles defined

96. Chronic HCV progresses silently because?

(a) Early infection is asymptomatic

(b) Always symptomatic

(c) Severe early symptoms

(d) Immediate liver failure

97. Hypertensive pregnancies risk?

(a) No complications

(b) Better outcomes

(c) Skin rash only

(d) Eclampsia

98. Sharing razors can transmit HIV due to?

(a) Sweat exposure

(b) No risk

(c) Blood contamination on razors

(d) Tears exposure

99. Efficient resource use means?

(a) Ignoring workflow

(b) Avoiding waste and repetition

(c) Spending without control

(d) Reducing staff only

100. High STI prevalence increases HIV spread because?

(a) STIs remove viral receptors

(b) STIs boost immunity

(c) STIs prevent HIV entry

(d) STIs increase viral load and mucosal susceptibility

1. **A farmer presents with a deep contaminated wound from soil. He never received vaccination.**
    a. Classify tetanus risk of this wound: **High risk** (as tetanus spores reside in soil and typically enter through deep puncture wounds or crush injuries)
    b. Outline **post-exposure prophylaxis: Wound cleaning, debridement, antibiotics, tetanus toxoid booster, and** Human Tetanus Immune Globulin **(HTIG).**
    c. List **primary prevention measures in community:**
    Health education on the importance of complete immunization,
    Promotion of safe wound care,
    Environmental sanitation,
    Community development,
    Mechanization of agriculture, & Reduction of exposure to animal waste.
2. **A neonate develops inability to suck and generalized spasms 7 days after home delivery by an untrained birth attendant.**
    a. Identify **risk factors for neonatal tetanus: Born to a non-immune mother and infection through the umbilical stump cut with unsterile instruments.**
    b. List **preventive measures at community level: Immunization of women of childbearing age**, **skilled attendance at delivery**, and **strict aseptic care** of **the umbilical stump**.
    c. Define **high-risk groups: Infants, the elderly, and unvaccinated persons.**
3. **A construction worker sustains repeated minor injuries but never seeks medical care.**
    a. Explain **mode of transmission and reservoir: Transmitted** through the **contamination of wounds with tetanus spores**. The **primary reservoirs** are **soil** and the **intestinal tracts of humans and animals.**
    b. Identify **occupational risk groups: Agricultural workers** (due to soil/animal waste exposure), **Farmers**, **construction workers, Gardeners, Laborers.**
    c. List **health education messages for prevention:** Emphasize the **importance of complete immunization** and the **promotion of safe wound care.**
4. **A 30-year-old man with unknown immunization history presents with a clean surgical wound.**
    a. Differentiate **management of clean vs contaminated wounds:**

Clean wounds	If immunization history is **unknown** or **incomplete**, a **tetanus toxoid booster is indicated. Tetanus immune globulin (TIG) is *not* indicated.**
Contaminated wounds	If immunization history is **unknown** or **incomplete**, **both tetanus toxoid (for active immunity) and TIG (for passive immunity) are indicated.**

b. State **role of active vs passive immunization**

Active immunization	(Tetanus toxoid) persists for at least 10 years.
Passive immunization	(TIG or equine antitoxin) provides temporary protection specifically for post-exposure prophylaxis.

c. List **secondary prevention measures:**

- Prompt notification to health authorities.
- Administration of TIG and antibiotics (e.g., metronidazole).
- Airway management and supportive care.
- Initiation of active immunization during treatment (since recovery does not confer immunity).

5. A child bitten by a stray dog with unknown vaccination status

a. Classify exposure: Bite route (virus-laden saliva introduced into a wound).

b. Outline **PEP steps:**

- Wound management: Immediate and thorough cleaning with soap and water.
- *Specific immunologic protection*.
- Rabies immune globulin (HRIG): Infiltrate half around the wound and give the rest IM.
- Rabies vaccine (HDCV): Administer a course of vaccine (e.g., on days 0, 3, 7, 14, and 28) to elicit active immunity.

c. List **high-risk groups for pre-exposure prophylaxis:**

- Veterinarians.
- Laboratory staff working with rabies virus.
- Dog handlers, hunters, and field naturalists.
- International travelers to rabies-endemic areas.

6. A veterinarian frequently handles animals in rabies endemic area.

a. Identify **high-risk occupations:** Veterinarians, lab staff, dog handlers, hunters, and field naturalists.

b. Describe **pre-exposure prophylaxis schedule:** Three 1.0 ml intramuscular doses on days 0, 7, and 21 or 28.
Booster doses if risk continues

c. List **community control measures:** Registration and immunization of all dogs and cats, euthanasia of stray animals, and active surveillance.

7. A man bitten by a vaccinated domestic dog that can be observed.

a. Describe **management based on animal observation:**

- If the domestic dog is healthy and can be observed for 10 days, PEP may be withheld.
- If the animal remains healthy throughout the observation period, it is not considered rabid at the time of the bite.

b. List **criteria for stopping PEP**: PEP can be stopped if the dog is proven to be healthy

through observation, or if a laboratory test confirms the animal is not rabid.

c. Explain **role of surveillance:** Active surveillance involves laboratory testing (FA testing) on all wild and domestic animals involved in human exposures or suspected of having rabies.

8. Several rabies cases reported in a rural village.

a. Describe **epidemiological triad of rabies:**

- **Agent** (Rabies virus/Lyssa virus type I),
- **Host/Reservoir** (warm-blooded animals like dogs, foxes, and humans),
- **Environment** (enzootic areas).

b. List **community-level prevention strategies:**

- Registering/immunizing pets,
- stray animal control,
- active surveillance.

c. Identify **modes of transmission:**

Bite route (virus-laden saliva),

Non-Bite Route (corneal transplants from infected donors or respiratory transmission in occupational settings).

9. A community shows increased cases of jaundice linked to contaminated water.

a. Identify **type of hepatitis likely involved:** Hepatitis A virus (and potentially E).

b. Explain **mode of transmission:** Spreads through contaminated raw produce, seafood, or water via infected food handlers.

c. List **preventive measures:** Safe water, sanitation, & food hygiene.

10. A healthcare worker exposed to hepatitis B positive blood.

a. Outline **post-exposure prophylaxis:** single dose of HBIG and the hepatitis vaccine.

b. Identify **high-risk groups:** Physicians, healthcare/lab personnel, hemodialysis patients, IV drug abusers, and recipients of blood/organs.

c. List **primary prevention strategies:** Routine immunization, proper sterilization of needles/syringes, and screening of all donated blood.

11. A chronic hepatitis B carrier works as a food handler.

a. Explain **role of carriers in epidemiology:** Carriers harbor the organism without clinical symptoms, spread the infection, and maintain disease endemicity.
(Note: Hep B is transmitted via body fluids, not the fecal-oral route).

b. List **control measures:** Education, immunization of contacts, and early medical management.

c. Describe **screening strategies:** Testing for Hepatitis B surface antigen (HBsAg) in serum.

12. A newborn delivered to hepatitis B positive mother.

a. Outline **neonatal prophylaxis:** single dose of HBIG (0.5 ml IM) & the first dose of the vaccine concurrently within 12 hours of birth.

b. Explain **vertical transmission:** Perinatal transmission is common, exceeding a 70% transmission rate if the mother is HBeAg positive.

c. List **immunization strategies:** Routine infant immunization starting at birth, followed by doses at 2, 4, and 6 months.

13. Outbreak of typhoid linked to street food vendor.

a. Explain **role of carriers:** Carriers appear healthy but harbor organisms, acting as a dangerous and hard-to-detect source of infection.

b. Identify **mode of transmission:** Ingestion of food or water contaminated by the feces and urine of patients or carriers.

c. List **control measures:** Scrupulous cleanliness in food preparation, fly control, and excluding carriers from handling food.

14. A chronic typhoid carrier works in food service.

a. Define **carrier types:** Chronologically (incubatory, convalescent, contact), by duration (temporary, chronic), and by route of exit (urinary, intestinal, biliary).

b. Explain **public health importance:** They form a "submerged iceberg" of infection, maintaining endemicity and spreading disease freely.

c. List **preventive strategies:** Exclude carriers from food handling until local regulations are met (often requiring 3 consecutive negative cultures from feces and urine).

15. A rural area lacks safe water and sanitation.

a. Identify **risk factors for typhoid:** Ingestion of contaminated water and food.

b. Explain **environmental determinants:** The physical environment (unsafe water, poor sanitation) & socio-economic factors (lack of infrastructure) facilitate the fecal-oral transmission cycle of *S. Typhi*. The organism spreads from human feces to water or food and then to a new host.

c. List **primary prevention measures**: Purifying/chlorinating public water, proper human feces disposal, and boiling water/milk.

16. Family members of a confirmed typhoid case.

a. Describe **contact management:** Immunization of household members and nursing contacts.

b. Outline **chemoprophylaxis role:**
Chemoprophylaxis is not routinely recommended for family contacts. However, in some instances, it may be used for high-risk contacts.
(fluoroquinolones like ciprofloxacin) for *treatment* of cases, but emphasizes vaccines for *prevention* in contacts.

c. List **health education measures:** Instruct on personal hygiene, specifically handwashing after defecation and before preparing food.

17. Mention Modifiable Risk Factors of CAD (mention 6)

Smoking, hypertension, high serum cholesterol (>220 mg/dl), diabetes mellitus, obesity, and physical inactivity.

18. What are the Dietary Changes for prevention of CAD

Limit saturated fats to less than 10% of total energy,
Keep average cholesterol intake below 300 mg/day,
Maintain serum cholesterol under 200 mg/dl,

Proportionately increase carbohydrates.

19. Mention 6 of the Primary Prevention measurement for prevention of DM

1. **Improvements in nutritional habits (avoiding sweets and fatty foods, high intake of dietary fibers).**
2. **Maintenance of body weight by doing moderate exercise.**
3. **Correction of obesity.**
4. **Avoiding overnutrition and alcohol.**
5. **Changing lifestyle (regular exercises).**
6. **Avoidance of oral contraceptives and steroids.**

20. Mention 5 Chronic Infections that can cause cancer (name the infection and cancer)

1. **Hepatitis B Virus (HBV):** Liver cancer (hepatocellular carcinoma).
2. **Hepatitis C Virus (HCV):** Liver cancer.
3. **Human Papillomavirus (HPV):** Cervical cancer.
4. **Helicobacter pylori:** Stomach cancer.
5. **Schistosoma haematobium:** Urinary bladder cancer.

21. Outline the role of diet and nutrition in cancer

- About 30% of cancers are linked to diet.
- **Western diet** (high in fat, red/processed meat) increases risk of colorectal, breast, prostate, and endometrial cancers.
- **Fruits and vegetables** are protective, decreasing risk of digestive tract cancers.
- **High salt/preserved foods** increase risk of gastric cancer.
- **Protective micronutrients** include vitamins A, C, E, and folate.

22. Enumerate 6 warning signs that can be noticed in self-examination for early detection of cancer

1. A **lump or hard area** in the **breast** or **testes**.
2. A **change in a wart or mole.**
3. A **persistent change** in **digestive or bowel habits.**
4. **Persistent indigestion** or **difficulty in swallowing.**
5. **Persistent cough or hoarseness.**
6. **Blood loss from any natural orifice.**

23. Enumerate the 8 Elements of Primary Health Care (i.e. Components)

1. **Health Education** concerning prevailing health problems and the methods of preventing & controlling them.
2. **Promotion of food supply** and **proper nutrition.**
3. **Provision of comprehensive maternal and childcare.**
4. **Immunization of children against major communicable diseases.**
5. **Prevention and control of locally endemic diseases.**
6. **Provision of adequate supply of safe water.**
7. **Appropriate treatment** of **common diseases and injuries.**
8. **Provision of essential drugs.**

24. Mention the 4 PHC principles

Equity, Appropriate technology, Multi-sectoral approach, and Community participation.

25. Mention Objective of health education:

- **Promote healthy lifestyles** (nutrition, exercise, smoking cessation).
- **Prevent disease through awareness of risk factors and screening.**
- **Improve self-care in chronic conditions.**
- **Enhance compliance with treatment and follow-up.**
- **Empower families to take responsibility for their health.**
- **Support community health goals** (vaccination, maternal-child health, environmental safety).

26. Enumerate Communication Barriers

Social/cultural gaps between communicator and participants,
limited receptiveness,
negative attitudes,
contradictory messages,
insufficient emphasis by health professionals.

27. What are the Advantages of personal approach

- **More effective and impressive: educator directly faces the recipients.**
- **Recipients participate in discussion (interaction between sender and recipient).**
- **Valuable to discuss confidential matters or problems (e.g., family planning).**
- **Personal approach can influence KAP (Knowledge, attitude, and practice) of recipients.**

28. What are the Criteria for Healthful Housing & Standards

- **Adequate living space** – prevent overcrowding.
- **Safe indoor temperatures** – avoid extremes of heat and cold.
- **Structural safety** – minimize injury hazards.
- **Accessibility** – suitable for people with disabilities.
- **Sanitation & water supply** – safe drinking water, hygienic waste disposal.
- **Ventilation & air quality** – reduce indoor pollutants, ensure airflow.
- **Protection from vectors & chemicals** – minimize exposure to pests and toxic substances.

29. What are the Global effects due to air pollution

Acid rain, the greenhouse effect & global warming, ozone layer depletion, radioactive disasters, and dioxin pollution.

30. How to prevention & Control of water pollution

Strict industrial regulation (treating wastewater),
Safe agricultural practices (reducing runoff),
Proper waste management,
Community awareness/action,
Global cooperation.

1. Post-neonatal mortality refers to deaths after the neonatal period but before the first birthday. Which age range does this cover?

A. 29 days to 1 year
B. Birth to 7 days
C. 5–15 years
D. 1–4 years

2. If 300 still births and 15,000 live births occur, what is the denominator for Still Birth Rate?

A. 300
B. 1,000,000
C. 15,000
D. 15,300

3. Age Specific Death Rate focuses on mortality within a defined age group. What denominator is required for its calculation?

A. Total deaths
B. Mid-year population of same age group
C. Infant deaths
D. Live births

4. A country has 300 maternal deaths and 2,000,000 women aged 15–49. What is the maternal mortality rate?

A. 30 per 100,000
B. 15 per 100,000
C. 300 per 100,000
D. 150 per 100,000

5. During an outbreak of food poisoning among school children. Of 1000 school children ate the school meals, there were 100 cases of food poisoning reported. The primary attack rate (cumulative incidence rate) equals:

A. 5 cases of food poisoning per 100 school children (5%).
B. 10 cases of food poisoning per 100 school children (10%).
C. 1 case of food poisoning per 100 school children (1%).
D. 12 cases of food poisoning per 100 school children (12%).

6. A country with 1% annual growth rate will double its population in:

A. 23 years

B. 50 years

C. 35 years

D. 70 years

7. Country A (population 2,000,000) recorded 18,000 deaths in one year. Among 200,000 elderly (≥65 years), 8,000 deaths occurred; among the remaining 1,800,000 people <65, there were 10,000 deaths. What is the ASDR for ≥65 years per 1,000?

A. 20

B. 30

C. 40

D. 10

8. WHEN THE NUMBER OF EDUCATED FEMALES IS EXPRESSED TO TOTAL FEMALES PRESENT IN A VILLAGE. IT IS KNOWN AS:

A. Cumulative frequency

B. Proportion

C. Rate

D. Ratio

9. In a dormitory, 36 new influenza cases occurred among 100 susceptible contacts. What is the secondary attack rate?

A. 30%

B. 24%

C. 36%

D. 18%

10. A 70-year-old woman with long-standing hypertension and atrial fibrillation suffers a stroke. She is bedridden for weeks, develops aspiration pneumonia, and dies from sepsis. What is the leading cause of death in this case?

A. Stroke

B. Hypertension

C. Aspiration pneumonia

D. Sepsis

11. Infant Mortality Rate is a key indicator of child health, measuring deaths under one year of age. Which numerator is used?

A. Still births

B. Maternal deaths

C. Infant deaths under 1 year

D. Neonatal deaths

12. Egypt is currently in which stage of demographic transition?

A. Third

B. First

C. Second

D. Fourth

13. Country A (population 2,000,000) recorded 18,000 deaths in one year. Among 200,000 elderly (≥65 years), 8,000 deaths occurred; among the remaining 1,800,000 people <65, there were 10,000 deaths. Why might CDR be misleading?

A. Does not adjust for age

B. Does not include infant mortality

C. Ignores gender

D. Changes monthly

14. If 500 infants die out of 15,000 live births in a year, what is the Infant Mortality Rate?

A. 33.3 per 1000

B. 50 per 1000

C. 30 per 1000

D. 3.3 per 1000

15. A census error occurs due to untrained data collectors. What type of error is this?

A. Geometric error

B. Error in data

C. Arithmetic error

D. Error in numbers

16. If 5500 maternal deaths occur with 15,000 live births, what is the Maternal Mortality Ratio?

A. 36.7 per 100,000

B. 367 per 100,000

C. 3670 per 100,000

D. 36.7 per 1000

17. If a country has a Gross Reproductive Rate of 0.8, what does this imply?

A. Population is unaffected

B. Population is stable

C. Population is declining over time

D. Population is growing rapidly

18. A country in the second stage of demographic transition shows declining death rates but stable birth rates. What is the result?

A. Stable growth

B. Rapid population increase

C. Declining population

D. Stationary population

19. Population study shows that it has five different stages. Certain county is currently having decreased death rate and an increased birth rate, so we will place this county in:

A. Stage IV

B. Stage I

C. Stage III

D. Stage II

20. Country A (population 2,000,000) recorded 18,000 deaths in one year. Among 200,000 elderly (≥65 years), 8,000 deaths occurred; among the remaining 1,800,000 people <65, there were 10,000 deaths. What is the crude death rate (CDR) per 1,000?

A. 9

B. 7

C. 12

D. 5

21. A country with a convex border and flat apex in its pyramid is:

A. Declining

B. Developing

C. Developed

D. Expanding

22. Which rate is used during outbreaks?

A. Incidence

B. Case Fatality Rate

C. Prevalence

D. Attack rate

23. Which rate is used during outbreaks?

A. Attack rate

B. Prevalence

C. Case Fatality Rate

D. Incidence

24. New Zealand, a developed country, has 23% of its population less than 15, and 12% over 65. This makes 65% between 15 and 64. Total dependency ratio:

A. 18.5

B. 53.8

C. 35.4

D. 16.9

25. THE NUMERATOR FOR THE INCIDENCE RATE IS:

A. Number of severe cases

B. Number of old cases

C. Number of new cases

D. Number of population at risk

26. A public health campaign aims to reduce the rate of new HIV infections in a region. Which morbidity indicator is the most appropriate primary metric for immediately evaluating the success of this specific prevention program?

A. Incidence rate

B. Years lived with disability

C. Point prevalence rate

D. Case fatality rate

27. Considering the population pyramid, the left side of pyramid is used to show

A. Mortality rate

B. Birth rate

C. Male population

D. Female population

28. A developed country shows a convex border and obtuse apex in its population pyramid. What does this reflect?

A. High mortality

B. High elderly population

C. Declining fertility

D. High birth rate

29. A population pyramid with wide base and tapering edges is typical of:

A. Stationary countries

B. Declining countries

C. Developed countries

D. Developing countries

30. A 45-year-old man with poorly controlled type 1 diabetes develops diabetic ketoacidosis (DKA). During hospitalization, he develops acute renal failure and dies from cardiac arrhythmia due to severe electrolyte imbalance. What is the underlying cause of death in this case?

A. Type 1 diabetes mellitus

B. Diabetic ketoacidosis

C. Cardiac arrhythmia

D. Electrolyte imbalance

31. If 50 new cases of measles occur in a population of 10,000 during one month, the incidence rate per 1,000 is:

A. 50.0

B. 500.0

C. 5.0

D. 0.5

32. A clinic surveys 3,200 adults on a single day and identifies 256 existing cases of hypertension. What is the point prevalence per 1,000 population?

A. 100 per 1,000

B. 80 per 1,000

C. 60 per 1,000

D. 40 per 1,000

33. If a community has 100 existing cases of diabetes in a population of 5,000, the prevalence per 1,000 is:

A. 200

B. 20

C. 10

D. 25

34. Which rate includes stillbirths and neonatal deaths in first week?

A. Infant Mortality Rate

B. Perinatal mortality rate

C. Maternal Mortality Rate

D. Neonatal mortality rate

35. A country's dependency ratio is 60%. What does this mean?

A. 60% of population is elderly

B. 60 working-age per 100 dependents

C. 60% of population is children

D. 60 dependents per 100 working-age people

36. If a community has 100 existing cases of diabetes in a population of 5,000, the prevalence per 1,000 is:

A. 20

B. 25

C. 10

D. 200

37. WHEN THE PREVALENCE, INCIDENCE AND AVERAGE DURATION OF THE DISEASE ARE CONSTANT, THE PREVALENCE RATE IS INCREASED BY:

A. Decrease in the average duration of the disease

B. increased survival and prolonged duration of the disease

C. Rapidly fatal disease.

D. Decrease in incidence rate

38. Neonatal Mortality Rate focuses on deaths occurring very early in life. Up to how many days after birth does it apply?

A. 28 days

B. 7 days

C. 1 year

D. 6 months

39. Perinatal mortality is a combined measure of early deaths. Which events are included?

A. Neonatal deaths only

B. Maternal deaths

C. Infant deaths under 1 year

D. Still births + deaths under 7 days

40. A newly introduced, highly effective treatment for a non-fatal chronic disease significantly extends the average life span of patients by 10 years, but does not cure the disease. What is the most likely long-term effect on the incidence rate (incidence rate) and the prevalence rate (pr) of this disease in the population?

A. Both stable

B. Incidence decreases; prevalence decreases

C. Incidence increases; prevalence stable

D. Incidence stable; prevalence increases

41. AN EXPERT IN THE FIELD OF PUBLIC HEALTH IS REQUIRED TO ESTIMATE THE MAGNITUDE OF A HEALTH PROBLEM. WHICH RATE WOULD HE CALCULATE FOR THIS?

A. Prevalence

B. Case fatality

C. Incidence

D. Proportionate mortality

42. A population pyramid with constant percentages over time is:

A. Declining

B. Constrictive

C. Expansive

D. Stationary

43. If 450 infants die between 29 days and 1 year out of 15,000 births, what is the Post-neonatal Mortality Rate?

A. 3 per 1000

B. 33 per 1000

C. 30 per 1000

D. 45 per 1000

44. A 58-year-old man with a history of chronic alcohol use and cirrhosis develops esophageal variceal bleeding. Despite transfusions, he dies from hypovolemic shock. What is the underlying cause of death in this case?

A. Hypovolemic shock

B. Esophageal variceal bleeding

C. Chronic alcohol use with cirrhosis

D. Blood transfusion complications

45. Country A (population 2,000,000) recorded 18,000 deaths in one year. Among 200,000 elderly (≥65 years), 8,000 deaths occurred; among the remaining 1,800,000 people <65, there were 10,000 deaths. What is the ASDR for <65 years per 1,000?

A. 8

B. 5.56

C. 12

D. 2.5

46. In a dormitory, 36 new influenza cases occurred among 100 susceptible contacts. What is the secondary attack rate?

A. 36%

B. 30%

C. 18%

D. 24%

47. In a standard population pyramid the age range of each bar is:

A. 20 years

B. 15 years

C. 10 years

D. 5 years

48. If 50 new cases of measles occur in a population of 10,000 during one month, the incidence rate per 1,000 is:

A. 5.0

B. 0.5

C. 50.0

D. 500.0

49. A NUMERICAL EXPRESSION WHICH INDICATES THE RELATIONSHIP IN AMOUNT OR SIZE BETWEEN TWO PARTS IS CALLED:

A. PERCENTAGE

B. RATE

C. PROPORTION

D. RATIO

50. A population pyramid shows a broad base and narrow apex. What does this indicate?

A. Low birth rate, high elderly population

B. High birth rate, low elderly population

C. Stable population

D. Declining population

51. A population of 1,000,000 experiences 10,000 deaths in a year. Using the formula for Crude Death Rate, what is the result?

A. 0.1 per 1000

B. 100 per 1000

C. 10 per 1000

D. 1 per 1000

52. If a country has a Gross Reproductive Rate of 0.8, what does this imply?

A. Population is growing rapidly

B. Population is unaffected

C. Population is declining over time

D. Population is stable

53. A 72-year-old woman with chronic obstructive pulmonary disease (COPD) develops pneumonia. Despite treatment, she dies from respiratory failure. What is the underlying cause of death in this case?

A. Hypoxemia

B. Respiratory failure

C. Pneumonia

D. COPD

54. With 50 neonatal deaths among 15,000 live births, what is the Neonatal Mortality Rate?

A. 33.3 per 1000

B. 50 per 1000

C. 3.3 per 1000

D. 30 per 1000

55. A census in Egypt is conducted every 10 years. Which method is used?

A. De facto

B. Graphic

C. De jure

D. Arithmetic

56. PMR is useful for analyzing mortality patterns. What does it help identify?

A. Migration

B. Leading causes of death

C. Fertility trends

D. Population growth

57. If 50 new cases of measles occur in a population of 10,000 during one month, the incidence rate per 1,000 is:

A. 0.5

B. 5.0

C. 500.0

D. 50.0

58. Still Birth Rate requires both live births and still births in the denominator. Which combination is correct?

A. Live births only

B. Infant deaths

C. Live births + still births

D. Mid-year population

59. If the ASFR for women aged 20–24 is 120 per 1,000, how many births are expected from 2,000 women in that age group?

A. 300

B. 200

C. 240

D. 120

60. Country A (population 2,000,000) recorded 18,000 deaths in one year. Among 200,000 elderly (≥65 years), 8,000 deaths occurred; among the remaining 1,800,000 people <65, there were 10,000 deaths. What is the ASDR for <65 years per 1,000?

A. 8

B. 2.5

C. 5.56

D. 12

61. A 3-day conference was attended by 600 people. Within 48 hours, 180 participants developed gastroenteritis. Investigators compared foods eaten by the sick individuals (cases) vs those who remained healthy (controls). What is the attack rate (%) for this outbreak?

A. 15%

B. 30%

C. 25%

D. 20%

62. Death certificates are divided into sections. What does Part I record?

A. Direct cause and antecedent causes

B. Other significant conditions

C. None of the above

D. Interval between onset and death only

63. A country with high dependency ratio faces which challenge?

A. More dependents per worker

B. More working-age population

C. Declining fertility

D. Stable growth

64. If a community has 100 existing cases of diabetes in a population of 5,000, the prevalence per 1,000 is:

A. 10

B. 20

C. 200

D. 25

65. The underlying cause of death is critical in certification. How is it defined?

A. Interval of death

B. Other significant condition

C. Immediate cause

D. Disease initiating morbid events

66. A country has 10,000 women aged 15–49 and 600 live births in a year. What is the General Fertility Rate (GFR)?

A. 6/1000

B. 0.6/1000

C. 60/1000

D. 600/1000

67. When a new treatment is developed that delays deaths but does not produce recovery from a chronic disease, which of the following will occur.

A. Prevalence of the disease will increase

B. Incidence of the disease will increase

C. Prevalence of the disease will decrease

D. Incidence of the disease will decrease

68. DURING AN OUTBREAK OF FOOD POISONING AMONG SCHOOL CHILDREN. OF 1000 SCHOOL CHILDREN ATE THE SCHOOL MEALS, THERE WERE 100 CASES OF FOOD POISONING REPORTED. THE PRIMARY ATTACK RATE (CUMULATIVE INCIDENCE RATE) EQUALS:

A. 1 case of food poisoning per 100 school children (1%).

B. 5 cases of food poisoning per 100 school children (5%).

C. 12 cases of food poisoning per 100 school children (12%).

D. 10 cases of food poisoning per 100 school children (10%).

69. A population pyramid with a narrow base and wide top indicates:

A. Expanding population

B. Stationary population

C. Declining population

D. Stable growth

70. A community has a population of 180,000 people. During the last year: • Total deaths: 1,200 • Number of heart disease cases: 4,500 • Deaths due to heart disease: 300 • Deaths due to all other causes: 900 What is the Proportionate Mortality Rate (PMR) for heart disease?

A. 20%

B. 25%

C. 12%

D. 30%

71. Country A (population 2,000,000) recorded 18,000 deaths in one year. Among 200,000 elderly (≥65 years), 8,000 deaths occurred; among the remaining 1,800,000 people <65, there were 10,000 deaths. What is the ASDR for ≥65 years per 1,000?

A. 30

B. 10

C. 40

D. 20

72. Case Fatality Rate is an important measure in epidemiology. What risk does it predict?

A. Risk of dying if disease is contracted

B. Risk of contracting disease

C. Risk of infant death

D. Risk of maternal death

73. A country with high fertility and declining mortality is in which stage of demographic transition?

A. First

B. Second

C. Fourth

D. Third

74. Which rate is used during outbreaks?

A. Case Fatality Rate

B. Attack rate

C. Prevalence

D. Incidence

75. A country has a population of 20 million in 1990 and 40 million in 2000. Using the arithmetic growth method, what is the estimated population in 2004?

A. 50 million

B. 48 million

C. 52 million

D. 44 million

76. It is an incidence rate used only when the population is exposed to risk for a limited period of time such as during an epidemic.

A. Relative risk

B. Attack rate

C. Odds ratio

D. Prevelance rate

77. Egypt's population growth rate is 1.7%. What is the approximate doubling time?

A. 23 years

B. 70 years

C. 41 years

D. 60 years

Formative 2 practical ans

1. A physician is asked why reconstituted vaccines like BCG must be used quickly. What is the correct explanation?

A. They become stronger with time

B. They cause more side effects if delayed

C. They are less effective when injected IM

D. They lose potency after a few hours

2. A rural physician notices that many children in his community are not completing their vaccination schedule. Which of the following is the most important role he should perform to address this issue?

A. Register births and deaths

B. Offer school health services

C. Conduct health education sessions

D. Provide outpatient treatment

3. Mass media may not necessarily influence behaviors. What does it mostly provide?

A. Behavior change

B. Full personal counseling

C. Knowledge and information

D. National policies

4. Destroying contaminated food targets:

A. Agent

B. Environment

C. Vector

D. Host

5. Personal approach includes face to face education. What is the main characteristic of this approach?

A. No interaction

B. Listening only

C. Direct contact and discussion

D. Use of mass media

6. Best study for small defined outbreak:

A. Cross-sectional

B. Cohort

C. Case-control

D. Ecological

7. Attack rate is used when:

A. Genetic disorders

B. Chronic disease

C. Limited exposure period

D. Long-term disease

8. A newborn is delivered normally and is stable. The mother asks when she should start breastfeeding. What is the recommended timing?

A. Within 6 hours

B. After colostrum stops

C. After 24 hours

D. Within 1 hour

9. Multiple peaks indicate:

A. No outbreak

B. Person-to-person spread

C. Lab error

D. Continuous exposure

10. Which situation is LEAST likely to be an outbreak?

A. Single sporadic case

B. Sudden rise in cases

C. Cases exceed expected

D. Cluster with excess numbers

11. Written report is important for:

A. Immediate control

B. Sampling

C. Legal documentation

D. Diagnosis

12. A point source outbreak shows:

A. Gradual rise

B. Continuous spread

C. Sharp peak

D. Multiple peaks

13. Isolation of cases reduces:

A. Exposure

B. Transmission

C. Immunity

D. Diagnosis

14. Failure to determine expected cases may lead to:

A. Lab error

B. Overestimation of RR

C. False outbreak assumption

D. Poor surveillance

15. Epidemic curve is best described as:

A. Pie chart

B. Histogram of cases over time

C. Table

D. Map

16. RR > 1 indicates:

A. Bias

B. Protective factor

C. No association

D. Increased risk

17. A child is weaned improperly and develops malnutrition. What is the likely cause?

A. Lack of energy-dense complementary foods

B. Early introduction of colostrum

C. Excessive breastfeeding

D. Use of pacifiers

18. Attack rate is a form of:

A. Incidence

B. Case fatality

C. Prevalence

D. Mortality

19. A mother interrupts her baby's feed after 5 minutes because she thinks the baby is full. What is the correct advice?

A. Stop feeding early

B. Use pacifier after 5 minutes

C. Allow baby to finish naturally

D. Switch to formula immediately

20. Final step in outbreak investigation:

A. Diagnosis

B. Communication

C. Treatment

D. Isolation

21. A spot map helps identify:

A. Incubation period

B. Clusters

C. Lab errors

D. Time trends

22. A mother with breast carcinoma asks if she can breastfeed. What is the advice?

A. Safe with precautions

B. Relative contraindication

C. Safe after 6 months

D. Absolute contraindication

23. Verifying diagnosis mainly ensures:

A. Prevention

B. No lab error

C. Case counting

D. Source identification

24. Hypothesis evaluation can be done using:

A. Analytic epidemiology

B. Treatment trials

C. Observation only

D. Lab tests only

25. Community participation strengthens primary health care services. Which group is important in this participation?

A. Shopkeepers

B. Non-governmental organizations (NGOs)

C. Foreign tourists

D. Comedians

26. Seminars are a two-way communication method used in group education. What makes seminars unique?

A. No questions allowed

B. Mainly for entertainment

C. Only one person speaks

D. Interaction between participants and instructor

27. Wide curve with long duration suggests:

A. Continuous exposure

B. Sporadic cases

C. No outbreak

D. Point source

28. Continued surveillance ensures:

A. Treatment success

B. Lab accuracy

C. Control effectiveness

D. Diagnosis

29. Person analysis helps identify:

A. Case definition

B. High-risk groups

C. Lab confirmation

D. Transmission mode

30. If peaks are separated by incubation period, this suggests:

A. Random variation

B. Food poisoning

C. Vector transmission

D. Propagated outbreak

31. A mother cannot breastfeed immediately due to surgery. What should be done?

A. Provide glucose water

B. Delay feeding until recovery

C. Express milk and feed newborn

D. Give formula only

32. Contradictory messages create communication barriers. Which situation illustrates a contradictory message?

A. A nurse giving clear instructions

B. A patient asking questions

C. An educator giving visual aids

D. A doctor telling a patient to rest while another says to exercise immediately

33. A health worker places food and drinks in the vaccine refrigerator. What is the risk?

A. Faster cooling of vaccines

B. Increased vaccine potency

C. Reduced vaccine potency due to contamination and temperature fluctuation

D. No effect on vaccines

34. Practical demonstrations are valuable in learning certain health skills. Which is an example?

A. Listening to the radio

B. Watching a TV show

C. Reading a newspaper

D. Demonstrating ORS preparation

35. Active surveillance differs from passive by:

A. Less accuracy

B. Investigator seeks cases

C. Lower cost

D. No reporting

36. A pandemic differs from an epidemic by:

A. Geographic spread

B. Mode of transmission

C. Duration

D. Severity

37. The FIRST step in outbreak investigation is:

A. Verify diagnosis

B. Develop hypothesis

C. Prepare for fieldwork

D. Identify cases

38. A mother asks about the effect of breastfeeding on fertility. What is correct?

A. Causes permanent infertility

B. Has no effect on fertility

C. Increases fertility immediately

D. Delays return to fertility

39. One goal of campaigns is influencing specific behaviors. Which health campaign objective is appropriate?

A. Reducing health education materials

B. Promoting seatbelt use

C. Increasing smoking rates

D. Encouraging refusal of vaccines

40. A child develops measles despite being vaccinated. Which of the following is the most likely cause?

A. Vaccine given at wrong age

B. Vaccine stored in sunlight

C. Vaccine administered without alcohol sterilization

D. Vaccine given orally instead of IM

41. Special surveillance is needed because routine may:

A. Be expensive

B. Overreport

C. Be biased

D. Miss cases

42. Campaigns are used for specific aims in health education. Which is an example of such a campaign?

A. Selling personal products

B. Promoting unhealthy habits

C. Increasing road accidents

D. Encouraging safety belt use

43. Health education uses various methods depending on goals and audience. Which method uses indirect teaching and keeps topics in the public eye?

A. Home visits

B. Personal approach

C. Mass communication

D. Role playing

44. Analytic epidemiology requires:

A. No comparison group

B. Only cases

C. Comparison group

D. Only controls

45. Mass communication is suitable for national health problems. Which of the following is an example?

A. Private counseling

B. Malnutrition awareness

C. A personal family dispute

D. A local shop opening

46. A health worker finds expired vaccines in the refrigerator. What should be done according to PHC guidelines?

A. Use them immediately before disposal

B. Remove them from the refrigerator

C. Store them in the door compartment

D. Mix with new vaccines to dilute effect

47. Immunization increases:

A. Susceptibility

B. Transmission

C. Exposure

D. Resistance

48. A major function of communication is directing the receiver toward behavior change. Which action reflects this function?

A. Confusing the audience

B. Stimulating thought and provoking questions

C. Reinforcing unhealthy habits

D. Limiting feedback

49. Epidemic curve represents:

A. Geographic spread

B. Risk factors

C. Time distribution of cases

D. Mortality trend

50. A nurse opens the refrigerator door frequently during vaccination. What is the likely consequence?

A. Vaccines remain potent

B. Vaccines freeze faster

C. Vaccines lose potency due to temperature fluctuation

D. Vaccines become contaminated

51. Understanding the attitude of recipients is important in health education. Why is this necessary?

A. To make communication more difficult

B. To ignore audience needs

C. To ensure messages are rejected

D. To tailor messages effectively

52. RR < 1 suggests:

A. Protective effect

B. Harmful exposure

C. Random error

D. No effect

53. Audiovisual media include tools used in education. Which of the following is an example of audiovisual media?

A. Radio only

B. Television

C. Newspapers

D. Posters only

54. Shock tactics in campaigns may use emotional imagery to influence behavior. What is an example mentioned in the lecture?

A. Broadcasting calm music

B. Showing peaceful landscapes

C. Showing cooking lessons

D. Displaying damaged cars on roadsides

55. Cohort study measures:

A. Relative risk

B. Sensitivity

C. Odds ratio

D. Prevalence

56. Curve shows multiple progressively taller peaks. What does this indicate?

A. Common source

B. Lab error

C. Person-to-person spread

D. Sporadic disease

57. High attack rate among exposed suggests:

A. Bias

B. Strong association

C. No association

D. Random error

58. In communication, the sender must design a message that attracts attention and interest. Which feature increases the effectiveness of a message?

A. Being specific and clear

B. Being contradictory

C. Being very long

D. Being vague

59. A mother asks why breast milk is better than formula. Which is correct?

A. Breast milk is contamination-free

B. Formula has more vitamins

C. Formula prevents infections

D. Formula prevents allergies

60. Establishing existence of outbreak requires:

A. Comparing observed vs expected cases

B. Counting deaths

C. Lab confirmation

D. Isolation of cases

61. The main purpose of a case definition is:

A. Treatment

B. Standardization

C. Isolation

D. Prevention

62. A gradually increasing curve over time suggests:

A. Common source

B. Propagated spread

C. Immunization effect

D. Lab error

63. Chemoprophylaxis is used for:

A. Isolation

B. Prevention

C. Diagnosis

D. Treatment

64. An epidemic curve shows a single sharp peak within one incubation period. What type of outbreak?

A. Continuous

B. Propagated

C. Endemic

D. Point source

65. Environmental investigation includes:

A. Treatment

B. Sampling food/water

C. Lab diagnosis

D. Patient interviews only

66. A mother asks about feeding frequency. What is correct?

A. Feeding once daily is enough

B. Scheduled feeding every 3 hours

C. Demand feeding is best

D. Only night feeding is needed

67. During a vaccination session, a nurse sterilizes the injection site with alcohol and immediately injects the vaccine. What is the likely consequence?

A. Vaccine potency is preserved

B. Vaccine may be damaged

C. No effect on vaccine efficacy

D. Child may develop immunity faster

68. Simulation exercises allow learners to practice realistic scenarios. What is their benefit?

A. They offer controlled learning environments

B. They remove the need for real practice

C. They avoid interaction

D. They eliminate feedback

69. A curve with steep rise and gradual decline indicates:

A. Propagated

B. Point source

C. Seasonal

D. Endemic

70. Asking patients about others with similar illness is:

A. Random sampling

B. Contact tracing

C. Passive surveillance

D. Screening

71. A baby is born and receives colostrum. Which of the following is a key benefit?

A. Causes dehydration

B. Delays gut development

C. Provides excess calories

D. Acts as first immunization

72. A mother asks why colostrum is sufficient despite small quantity. What is the reason?

A. High caloric density

B. Prevents uterine involution

C. Causes rapid weight gain

D. Contains excess water

73. Descriptive epidemiology includes:

A. Exposure, outcome, control

B. Time, place, person

C. Cause, effect, treatment

D. Lab, clinical, imaging

74. A child is given MMR vaccine at 12 months. What is the route of administration?

A. ID

B. Oral

C. IM

D. SC

75. Restricting cases by time helps to:

A. Control bias

B. Identify treatment

C. Link exposure period

D. Increase cases

76. A sudden spike followed by no cases indicates:

A. Continuous exposure

B. Single exposure event

C. Endemic pattern

D. Seasonal trend

77. A mother asks how to know if attachment is good. Which is correct?

A. Nose and chin touch breast

B. Mouth closed tightly

C. Baby's lips turned inward

D. Only nipple inside mouth

78. Descriptive epidemiology helps in:

A. Hypothesis generation

B. Vaccine development

C. Diagnosis

D. Treatment

79. A mother breastfeeds for a long duration. Which long-term benefit is correct?

A. Lower risk of arteriosclerosis

B. Increased maternal cancer risk

C. Higher risk of arteriosclerosis

D. Increased infant mortality

80. Propagated outbreaks are characterized by:

A. Single peak

B. Flat curve

C. Multiple waves

D. No trend

81. Mass communication is useful in keeping health issues constantly visible to the public. Which example fits this purpose?

A. TV campaign on accident prevention

B. Personal interview

C. Home visitation

D. Confidential family counseling

82. Hypothesis should include:

A. Treatment plan

B. Source and transmission

C. Lab methods

D. Cost analysis

83. A case definition must include:

A. Clinical + time/place/person

B. Only symptoms

C. Treatment protocol

D. Only lab findings

84. A disadvantage of mass communication is lack of feedback. What challenge does this cause?

A. Better interpersonal connection

B. Immediate behavior change

C. Increased interaction

D. Difficulty in assessing message understanding

85. A rural physician is conducting a health education session. Which of the following topics is most relevant to promotive care?

A. Provision of essential drugs

B. Treatment of endemic diseases

C. Immunization against measles

D. Proper nutrition and food supply

86. Community participation involves NGOs and influential groups in delivering health services. What is a major benefit of community participation?

A. Weakens service delivery

B. Reduces public involvement

C. Eliminates the need for health workers

D. Enhances health education delivery

87. Which best defines an epidemic?

A. Sudden increase above expected level

B. Worldwide spread

C. Seasonal variation

D. Constant disease presence

88. A flat prolonged curve suggests:

A. No outbreak

B. Point source

C. Propagated

D. Continuous exposure

89. Epidemic curve helps determine:

A. Case definition

B. Mode of transmission

C. Lab test

D. Treatment

90. A disease consistently present in a region is called:

A. Pandemic

B. Sporadic

C. Epidemic

D. Endemic

Practical formative 3

1. What does psychosomatic outcome from stress include?

A. Cataract

B. Epistaxis

C. Asthma

D. Dermatitis

2. What is the purpose of preplacement exam?

A. Diagnose chronic disease

B. Reduce salary

C. Reject healthy workers

D. Fit worker to appropriate job

3. A treatment plant uses Dorr Clarifier. What is its main advantage?

A. Enhances sedimentation efficiency

B. Removes viruses

C. Adds chlorine

D. Neutralizes acidity

4. Fish die in a canal after fertilizer runoff. Which process is most likely occurring?

A. Filtration

B. Sedimentation

C. Chlorination

D. Eutrophication

5. Radiation traces are found in water. Which standard is violated?

A. Physical

B. Radiological

C. Chemical

D. Biological

6. A laboratory worker exposed to nonionizing radiation complains of progressive visual opacity impairing vision. What radiation type most likely caused this?

A. Infrared radiation

B. Microwave

C. Laser

D. Ultraviolet radiation

7. A water sample contains 600 ppm solids. What does this exceed?

A. Maximum solids limit

B. Maximum hardness

C. Minimum chlorine requirement

D. Maximum nitrate limit

8. A worker involved with hammer drills presents with numbness and blanching of fingers. Which hazard caused this?

A. Radiation

B. Heat

C. Noise

D. Vibration

9. Microspora faeni is most associated with which of the following?

A. Farmer's lung

B. Silicosis

C. Asbestosis

D. Bagassosis

10. Hardness of 25 degrees is reported. What does this exceed?

A. Minimum chlorine requirement

B. Maximum solids limit

C. Maximum nitrate limit

D. Maximum hardness limit

11. Metal fume fever occurs most commonly in which of the following?

A. Welders

B. Farmers

C. Office employees

D. Drivers

12. A new factory worker presents with painful muscle spasms after his first day in a hot environment. Lab results show low sodium and chloride. Which heat disorder best explains this presentation?

A. Heat cramps

B. Heat stroke

C. Heat exhaustion

D. Heat syncope

13. What does isolation, as a safety measure, mean?

A. Reducing workforce

B. Using PPE only

C. Physically separating workers from hazards

D. Increasing ventilation

14. Neutral pH but high turbidity is found. Which standard is violated?

A. Radiological

B. Biological

C. Physical

D. Chemical

15. A water sample shows turbidity and odor. Which standard is violated?

A. Biological

B. Radiological

C. Physical

D. Chemical

16. Residual chlorine after 30 minutes is 0.2 ppm. What does this indicate?

A. Excess chlorine

B. Safe disinfection

C. Insufficient chlorine

D. No chlorine added

17. A village stores rainwater in tanks but residents complain of metallic taste. What is the most likely cause?

A. High turbidity

B. Rainwater dissolving metals

C. Excess chlorine

D. Protozoa contamination

18. Which of the following is correct about pneumoconiosis fibrosis?

A. Reversible

B. Curable

C. Progressive and permanent

D. Preventable but treatable

19. A plant reports bacterial reduction to 5% of original load after filtration. Which method was used?

A. Rapid sand filter

B. Slow sand filter

C. Reverse osmosis

D. Boiling

20. What does environmental monitoring includes?

A. Air sampling

B. Stool analysis

C. Blood pressure measurement

D. ECG

21. A chemical plant employee develops dermatitis, folliculitis, and chronic eczema. Which category of agents is most responsible?

A. Allergens

B. Ingestants

C. Asphyxiants

D. Irritants

22. A community experiences diarrhea outbreaks despite filtration. Which step was insufficient?

A. Intake

B. Sedimentation

C. Filtration

D. Disinfection

23. Clostridium welchi is detected in water. What does this suggest?

A. Viral infection

B. Recent contamination

C. No contamination

D. Remote contamination

24. From which Lead apron protects against?

A. Ionizing radiation

B. Microwave rays

C. UV radiation

D. IR radiation

25. Streptococcus faecalis is found in water. What does this indicate?

A. Safe water

B. Chemical pollution

C. Recent fecal contamination

D. Remote contamination

26. Pathogenic protozoa are detected. Which treatment is most effective?

A. Boiling only

B. Sedimentation only

C. Chlorination only

D. Rapid and slow sand filtration

27. A water sample shows 0.5 ppm nitrites. What does this indicate?

A. Ancient organic matter

B. Past pollution

C. Safe water

D. Recent and ongoing pollution

28. Worker returns after vacation and experiences chest tightness after first shift. Which disease?

A. Byssinosis

B. Farmer's lung

C. Weaver's cough

D. Mill fever

29. Which measure prevents bagassosis most effectively?

A. Using earplugs

B. Spraying 1% propionic acid

C. Increasing lighting

D. Antibiotics

30. Residents complain of unpleasant taste despite disinfection. Which factor is most likely responsible?

A. High turbidity

B. Low pH

C. Excess chlorine

D. Excess alum

31. Hardness due to calcium bicarbonates is found. What type of hardness is this?

A. Permanent

B. Radiological

C. Biological

D. Temporary

32. A worker exposed to lowgrade wet cotton develops greenish sputum, mild dyspnea, and headaches. Likely diagnosis?

A. Byssinosis

B. Weaver's cough

C. Bagassosis

D. Mill fever

33. A worker develops acute fever, chills, malaise, and dyspnea 6 hours after exposure to bagasse dust. Which disease fits?

A. Byssinosis

B. Silicosis

C. Farmer's lung

D. Bagassosis

34. Klebsiella aerogens is detected. What does this indicate?

A. Non-fecal coliform presence

B. Safe water

C. Fecal contamination

D. Viral contamination

35. A worker with asbestosis develops rapid onset pleural effusion and chest pain. Which malignancy is suspected?

A. Sarcoma

B. Lung adenocarcinoma

C. Squamous carcinoma

D. Mesothelioma

36. A water sample shows pH 5.5. Why is this unsuitable?

A. Too alkaline

B. Too turbid

C. Too acidic

D. Too hard

37. Pathogenic viruses are detected in water. Which statement is correct?

A. Viruses are acceptable in small amounts

B. Chlorination cannot remove viruses

C. Drinking water should be free from viruses

D. Viruses indicate chemical pollution

38. Engineers are designing intake pipes for a growing city. Where should they be placed to avoid pollution?

A. At the bottom of the river only

B. Downstream near factories

C. Upstream from the city and pollution sources

D. At the riverbank only

39. Which is the most important engineering measure for dust control in sandblasting?

A. Improve lighting

B. Replace sand

C. Add more workers

D. Reduce PPE

40. Bacterial count is 100 organisms/ml. Is this acceptable?

A. No, indicates radiation

B. No, too low

C. No, exceeds limit

D. Yes, within standard

41. What type of hearing loss is caused by chronic exposure to high noise?

A. Sensorineural loss

B. Mixed loss

C. No hearing loss

D. Conductive hearing loss

42. Why particles <0.1 μm are harmless?

A. High solubility

B. Brownian movement

C. Chemical inertness

D. Electrostatic charge

43. A worker in asbestos cement industry presents with basal crepitations and "groundglass" appearance on Xray. Which complication is highly associated?

A. Mesothelioma

B. Pulmonary embolism

C. Pneumonia

D. Raynaud's disease

44. What does health education in occupational settings include?

A. Supervising stock markets

B. Managing worker salaries

C. Training on proper PPE use

D. Housing allocation

45. Nitrites absent but nitrates at 0.8 ppm. What does this indicate?

A. Safe water

B. Past pollution

C. Excess chlorine

D. Recent pollution

46. A worker with silicosis develops sudden weight loss, night sweats, and productive cough. Which disease is now suspected?

A. Sarcoidosis

B. Asthma

C. Tuberculosis

D. Bronchitis

47. Albuminoid ammonia is measured at 0.05 ppm. What does this indicate?

A. Neutral pH

B. High chlorine levels

C. Safe water

D. Excess organic decomposition

48. In a worker exposed to Infra Red radiation, what is the primary ocular damage?

A. Retinal hemorrhage

B. Cataract

C. Conjunctivitis

D. Glaucoma

49. A miner working in dimly lit tunnels complains of progressive eye fatigue and intermittent oscillatory eye movements. What condition is most likely responsible?

A. Miner's nystagmus

B. Photophobia

C. Snow blindness

D. Keratitis

50. What does good housekeeping principle state?

A. Clean once daily

B. Cleaning is optional

C. Everything has a place and in its place

D. Workers clean only

51. Chlorine gas is added. After 30 minutes, residual chlorine is 0.15 ppm. Was this effective?

A. No, too low

B. No, chlorine should not be used

C. Yes, within safe range

D. No, too high

52. Alum is added during sedimentation. What is its primary function?

A. Form aluminum hydroxide floc to trap impurities

B. Neutralize acidity

C. Increase alkalinity

D. Kill bacteria directly

53. E. coli is found in water. What does this confirm?

A. Fecal contamination

B. Safe drinking water

C. Viral contamination

D. Chemical contamination

54. A miner's chest Xray shows eggshell calcification of hilar lymph nodes. What is the diagnosis?

A. Farmer's lung

B. Byssinosis

C. Silicosis

D. Asbestosis

55. Nitrate levels of 6 ppm are found in deep well water. What does this suggest?

A. Indicates recent pollution

B. Safe concentration

C. Indicates chlorine overdose

D. Exceeds recommended limit

56. Oxygen absorbed in 3 hours at 27°C is 5 ppm. What does this exceed?

A. Nitrite limit

B. Chlorine residual standard

C. Organic matter oxidation standard

D. Hardness limit

57. Coliform bacteria are found in 100 ml of water. What does this violate?

A. Chlorine residual standard

B. Hardness standard

C. Nitrate standard

D. Presumptive coliform standard

58. A factory worker develops severe hostility, depression, and absenteeism due to workplace conflicts. Which hazard category is implicated?

A. Chemical

B. Physical

C. Mechanical

D. Psychosocial

59. White finger disease results from which of the following?

A. Chemical burns

B. Vibration

C. Heat exposure

D. Radiation

60. A cotton worker with byssinosis has normal Xray but reduced FEV1. What is the exposure duration?

A. Weeks

B. Days

C. <1 year

D. >10 years

PROBLEM SOLVING

By

Prof. Fouad Metry Atia

PROBLEM (1)

- **In 2005, Pakistan crude birth rate was 36 births per 1000 population and the crude death rate was 9 deaths per 1000 population.**
  - **What was the population growth rate of the country in that year assuming no in and out migrations?**
  - **Calculate duplication time?**

ANSWER

- **a. Population Growth Rate (Natural Increase)**
- Natural Increase $=$ CBR $-$ CDR
  - $36 - 9 = 27$ per 1,000 population
  - $27/1000 = 0.027 = 2.7\%$
- **b. Duplication (Doubling) Time**
  - Doubling Time $= \dfrac{70}{\text{Growth Rate } (\%)}$
  - $\frac{70}{2.7} \approx 25.9$ years
  - Doubling time $\approx 26 years$

PROBLEM (2)

- **New Zealand, a developed country, has 23% of its population less than 15, and 12% over 65. This makes 65% between 15 and 64. Calculate**
  - **a. Young Dependency ratio**
  - **b. Old dependency ratio**
  - **c. Total dependency ratio**

ANSWER

- $\text{Dependency Ratio} = \frac{\text{Dependent Group}}{\text{Working-Age Population}} \times 100$
- **a. Young Dependency Ratio**
- $\frac{23}{65} \times 100 = 35.38$
- **b. Old Dependency Ratio**
- $\frac{12}{65} \times 100 = 18.46$
- **c. Total Dependency Ratio**
- $\frac{23+12}{65} \times 100 = \frac{35}{65} \times 100 = 53.85$

PROBLEM (3)

- **A city contain 100.000 people (45.000 males and 55.000 females) and 1000 people die per year (600 males and 400 females).There were 50 cases (40 males and 10 females) of lung cancer per year ,of whom 45 died (36 males and 9 females) . <u>Calculate:</u>**
  - **Crude mortality rate.**
  - **Sex specific mortality rates.**
  - **Lung cancer specific mortality rates.**
  - **Case fatality rate of lung cancer.**
  - **Proportionate mortality rate for lung cancer.**

ANSWER

- **a. Crude Mortality Rate (CMR)**
- $\text{CMR} = \frac{\text{Total deaths}}{\text{Total population}} \times 1000$
- $\text{CMR} = \frac{1000}{100000} \times 1000 = 10 \text{ per 1000 population}$
- **b. Sex-Specific Mortality Rates**
- **Male Mortality Rate**
- $\text{Male mortality} = \frac{600}{45000} \times 1000 = 13.33 \text{ per 1000 population}$
- **Female Mortality Rate**
- $\text{Female mortality} = \frac{400}{55000} \times 1000 = 7.27 \text{ per 1000 population}$

- **c. Lung Cancer Specific Mortality Rate**
- Cause-specific mortality rate $= \frac{\text{Deaths from lung cancer}}{\text{Total population}} \times 1000$
- $\frac{45}{100000} \times 1000 = 0.45$
- **d. Case Fatality Rate (CFR)**
- $\text{CFR} = \frac{\text{Deaths from lung cancer}}{\text{Lung cancer cases}} \times 100$
- $\frac{45}{50} \times 100 = 90\%$
- **e. Proportionate Mortality Rate (PMR) for Lung Cancer**
- $\text{PMR} = \frac{\text{Deaths from lung cancer}}{\text{Total deaths}} \times 100$
- $\frac{45}{1000} \times 100 = 4.5\%$

PROBLEM (4)

- In a district with an mid year population of 200,000 people, There were 40 cases of measles reported during 2004 of whom 10 died. On December 31.2004, There were a total of 120 registered cases of TB of whom 20 died. Calculate all possible mortality rates.

ANSWER

- **Cause-Specific Mortality Rates**
  - **A. Measles-specific mortality rate**
  - $\frac{10}{200000} \times 1000 = 0.05$
  - **B. TB-specific mortality rate**
  - $\frac{20}{200000} \times 1000 = 0.10$
- **Case Fatality Rates (CFR)**
  - **A. Measles CFR**
  - $\frac{10}{40} \times 100 = 25\%$
  - **B. TB CFR**
  - $\frac{20}{120} \times 100 = 16.67\%$

PROBLEM (5)

- **In a certain village with a population of 50,000 people there are 5000 deaths during the year ending in December 30.2000. These include 1500 deaths due to malaria in 10,000 people who had developed the disease in the village this year.**
  - **What was the crude death rate?**
  - **What was the case fatality rate for malaria?**

ANSWER

- **a. Crude Death Rate (CDR)**
  - $\text{CDR} = \frac{\text{Total deaths}}{\text{Total population}} \times 1000$
  - $\text{CDR} = \frac{5000}{50000} \times 1000$
  - $\text{CDR} = 100 \text{ per } 1000 \text{ population}$
- **b. Case Fatality Rate (CFR) for Malaria**
  - $\text{CFR} = \frac{\text{Deaths from malaria}}{\text{Malaria cases}} \times 100$
  - $\text{CFR} = \frac{1500}{10000} \times 100$
  - $\text{CFR} = 15\%$

PROBLEM (6)

- **The following data came from the annual report of certain locality in Sohag governorate 2003.**
  - **Total deaths 1000**
  - **Infant deaths 200**
  - **Maternal deaths 20**
  - **Post neonatal deaths 120**
  - **No. of L.B 3000**
  - **Total population 50,000**
  - **Female population of age group (15-49 years) 10.000**
- **Compute the following rates:**
  - **Rate of natural increase.**
  - **Maternal mortality rate.**
  - **Neonatal mortality rate.**

ANSWER

- **1. Rate of Natural Increase**
- Rate of Natural Increase $= \frac{\text{Births} - \text{Deaths}}{\text{Total Population}} \times 1000$
- $3000 - 1000 = 2000$
- $\frac{2000}{50000} \times 1000 = 40$ per 1000 population = **4% per year**
- **2. Maternal Mortality Rate (MMR)**
- $\text{MMR} = \frac{\text{Maternal deaths}}{\text{Female population 15–49}} \times 100000$
- $\text{MMR} = \frac{20}{10000} \times 100000 = 200/100000$
- **3. Neonatal Mortality Rate (NMR)**
- Neonatal deaths = Infant deaths − Post-neonatal deaths
- $200 - 120 = 80$
- $\text{NMR} = \frac{80}{3000} \times 1000 = 26.67$

PROBLEM (7)

- **A 70-year patient died with advanced hydronephrosis, which was complicated by uraemia, and there was a long history of retention of urine due to hypertrophy of the prostate.**
- **a. What is the direct cause of death**
- **b. Determine the underlying cause of death.**

ANSWER

- **Cause-of-Death Statement**
  - A 70-year-old man died with:
  - Advanced **hydronephrosis**
  - Complicated by **uraemia**
  - Due to long-standing **urinary retention**
  - Caused by **hypertrophy of the prostate**
- **a. Direct (Immediate) Cause of Death**
  - **Uraemia**
- **b. Underlying Cause of Death**
  - **Hypertrophy of the prostate (Benign prostatic hyperplasia)**

PROBLEM (8)

- **A 9-year old child with bronchopneumonia and he was receiving chemotherapy for treatment of leukemia. He is malnourished and shows other manifestations of bone marrow suppression such as anemia and bleeding tendency.**
  - **a. What is the direct cause of death**
  - **b. Determine the underlying cause of death.**

ANSWER

- **Case Summary**
  - A 9-year-old child had:
  - **Bronchopneumonia**
  - **Leukemia** (under chemotherapy)
  - **Malnutrition**
  - **Bone marrow suppression** (anemia + bleeding tendency)

- **a. Direct (Immediate) Cause of Death**
  - **Bronchopneumonia**
- **b. Underlying Cause of Death**
  - **Leukemia**

PROBLEM (9)

- **A patient was admitted to the hospital with multiple fractures and his condition deteriorated rapidly followed by traumatic shock then death. The persons accompanying him to the hospital said that a car struck him while he was walking.**
- **a. What is the direct cause of death**
- **b. Determine the underlying cause of death.**

ANSWER

- **Case Summary**
  - A patient was:
  - Struck by a **car while walking**
  - Sustained **multiple fractures**
  - Condition deteriorated → **traumatic shock** → **death**
- **a. Direct (Immediate) Cause of Death**
  - **Traumatic shock**
- **b. Underlying Cause of Death**
  - **Car accident (road traffic injury)**

Thank you